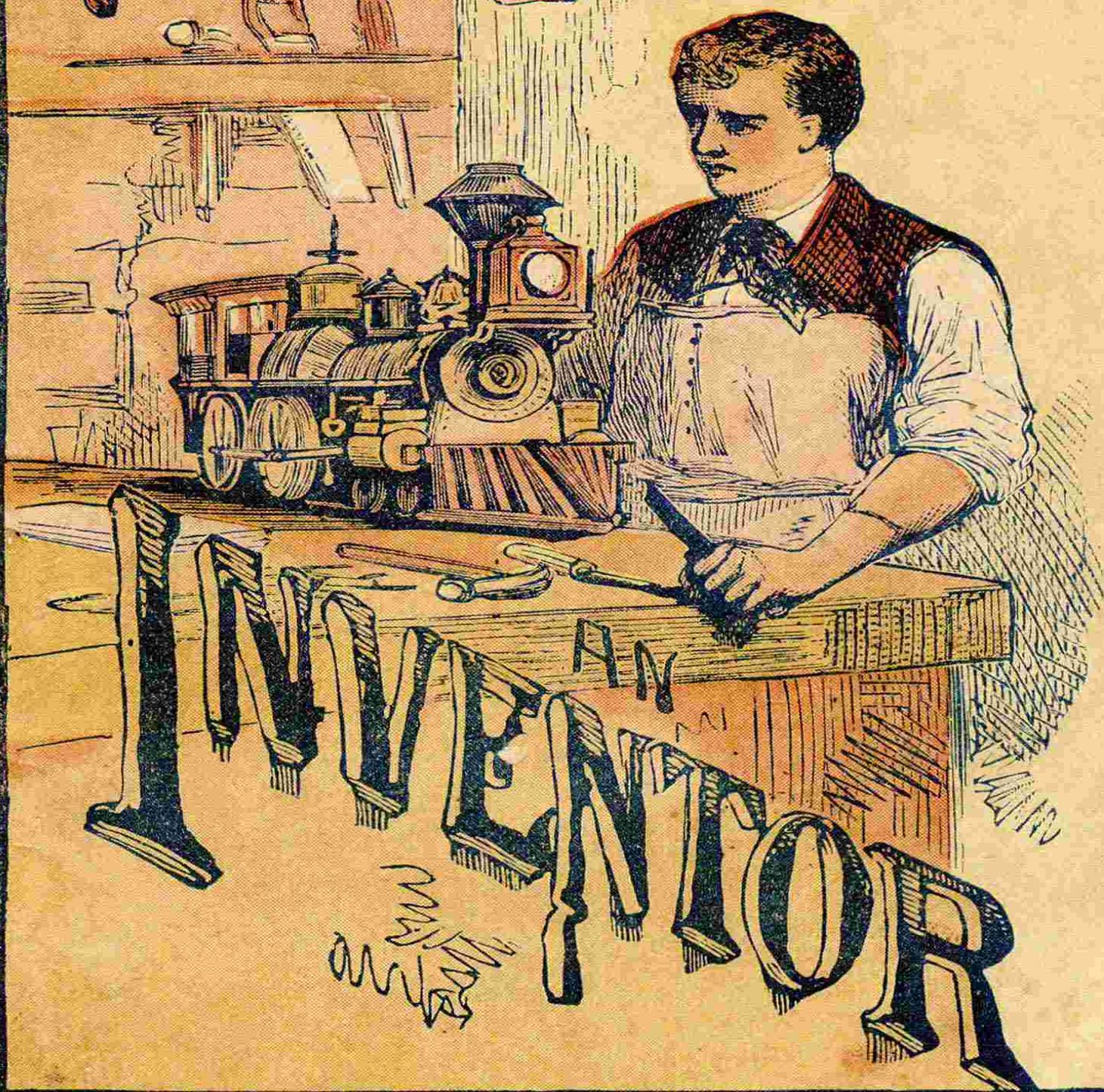
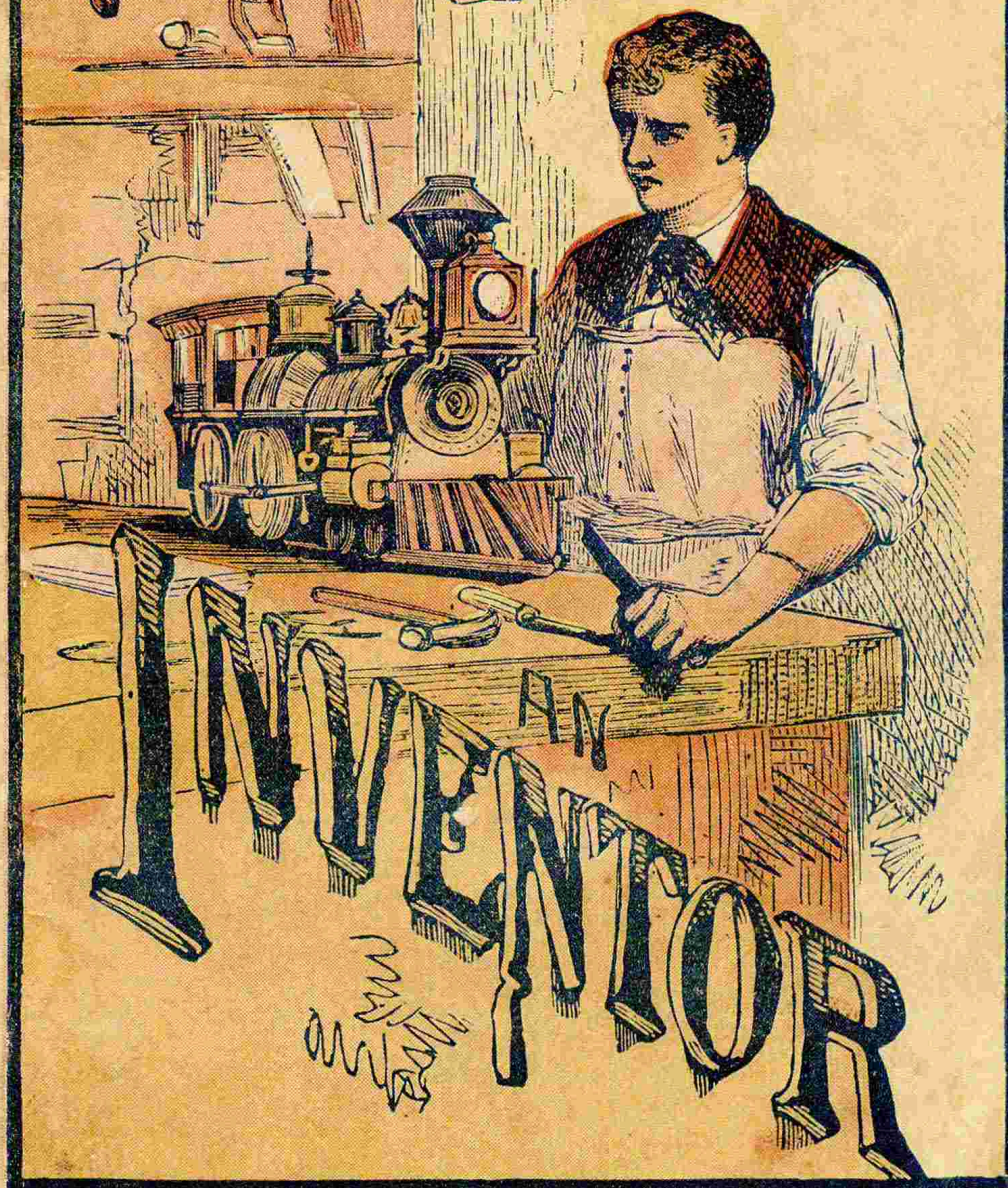


HOW TO BECOME



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Containing experiments in photography, hydraulics, galvanism and electricity, magnetism, heat, and the wonders of the microscope

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*** START OF THE PROJECT GUTENBERG EBOOK HOW TO
BECOME AN INVENTOR ***

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How to Become an Inventor.

CONTAINING

Experiments in Photography, Hydraulics,
Galvanism and Electricity,

MAGNETISM, HEAT,

AND THE

Wonders of the Microscope.

ALSO GIVING

Instruction in the Use of Tools

AND

OPTICAL INSTRUMENTS.

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How to Become an Inventor.

Nothing is more useful to a youth than to be able to do a little carpentering. To be handy with a chisel and saw, a nail and a hammer, saves many a dollar in the course of the year. If you call in a carpenter for a little work he is sure to spin out a “regular job.” I remember once buying some oak saplings, which cost me fifteen cents a stick; and wanting to build a summer-house, I required eight of them to be sawn through, so I applied to a carpenter, and the sticks were cut, but, to my astonishment, four dollars was charged for this little “job,” although the wood cost me only one dollar and thirty cents. I found out afterwards that the proper price for sawing would have amounted to about one dollar, so that three dollars profit was clapped on for the benefit of my experience. I just mention this to show my young friends that if they wish to make summer-houses for their gardens, cages for their birds, fowl-houses, rabbit-hutches, or boxes for their books, they must learn to make them for themselves. I shall therefore offer them a little advice upon “carpentering.”

THE SHOP AND BENCH.

Endeavor to procure some small outhouse, in which you may erect what is called a carpenter’s or joiner’s bench. These may very often be bought second-hand, or if not, can easily be procured at a reasonable rate. I am very particular in recommending a bench, as without it you will find many obstacles to your work. You must also provide yourself with a set of tools,—gimlets, hammers, planes, saws, gouges, files, nails, screws, and such articles of use.

The bench is composed of a platform or top, supported by four stout legs; supplied with a bench hook; this ought to be fitted in tight, so as to move up and down with a hammer only. The use of it is to keep any wood steady you may have to plane; the bench screw is used for keeping any wood firm and steady you may have to saw, which is to be put in the grip and screwed tight. Sometimes the edges of wood require to be planed, and then the wood is put in the grip or cheeks of the bench and held tight while you plane it. Make holes in the side of the bench, for the insertion of a movable pin to support the end of the board you have to plane or saw, which is not in the

screw. The height of your bench should be about 2 feet 8 inches. The common length is from 10 to 12 feet, and the breadth about 3 feet 6 inches.

The jack plane is the first to be used. It is about 17 inches in length, and is used to take the rough parts from a piece of wood. It should be held steadily by fixing the right hand at the handle, and the left over the top and side, and pushed forward on the wood, when the knife will take off a shaving which runs through the hole, and falls on the side. In using the plane the endeavor should be to take off a clean shaving, which is done by using the instrument uniformly and steadily over every surface to be planed.

There is another kind of plane, called the trying plane, having a double top or handle. It is used to regulate and smooth to a higher degree, the surface of the wood that had previously been smoothed from the rough by the jack plane. Its length is about 22 inches, and it is broader than the jack plane. There is another plane called the *long* plane, which is used for facing a piece of stuff, which it does with the greatest exactness; its length is about 2 feet 4 inches. There is also the joiner's plane, which is the longest of all the planes, being 30 inches long. But the most handy of the planes to the boy carpenter is the smoothing plane. It is the last plane used in joining, and gives the utmost degree of smoothness to the surface of a piece of finished work; it is about 7 inches in length, the sides of the stock are curved, and resemble in figure a coffin; it is used in a similar way to the other.

SAWS.

There are many kinds of saws, but the most useful one is what is called the "hand saw." It has a blade or plate about 28 inches long; the teeth of which are so formed as to allow you to cut the wood crossways as well as lengthways. The handle of the saw is made so as to allow a full yet free grasp of the hand, either for a pull or a thrust.

The *panel saw*. This saw has a plate nearly of the same size as a hand saw, and is used for cutting very thin boards, which the rough teeth of the hand saw would not cut through without breaking them.

The *tenon saw* is of a different shape to the others, and is made to cut across the grain of the wood so as to leave the ends nicely even, that it may fit to the piece it is joined to, which is called a shoulder, being that part which comes in contact with the fiber of the wood. To do this it requires that the

teeth should be much smaller, and they are therefore placed so close as eight or ten to the inch, according to the length of the blade.

The *dovetail saw*. There is another most useful saw it would be of advantage for the young carpenter to have, namely, the dovetail saw. It is about 9 inches long, and contains at least fifteen teeth in the inch. It is used for cutting the dovetails of boxes. Its plate is very thin, and it requires some care in using. It has a back for the purpose of strength, formed of a thin piece of brass or iron, let in so as to give the blade the requisite firmness necessary in using it.

The *compass saw*. The plate of this kind of saw is very narrow, and not more than one inch wide at the broadest part, gradually diminishing to about a quarter of an inch at the lower end. It is about 15 inches in length, and used for cutting a piece of wood into a circular form, and the plate being narrow allows it to follow the foot of the compass to a very small diameter.

The *keyhole saw*. The keyhole saw is much smaller than the above. It is used for cutting short curves, small holes, &c., such as a keyhole. The handle is the same form as that of the chisel, a small slit being cut through from end to end. It has a screw on one side, in order that the blade may be set to any length, according to the circumference of the hole to be cut.

THE SPOKE SHAVE.

This is a very useful tool. It is employed for smoothing the edges of round pieces, or other ends requiring to be shaved down. It is a narrow plane made of boxwood, and has generally a steel blade let into it to cut; it is used by taking hold of each end with a hand, and moving it to and fro over the wood to be shaved down.

STOCK AND BITS.

There are about thirty-six bits to a set, all of different shapes and sizes; but our young friends need not get quite so many; if they provide themselves with a couple of a medium size, this will be sufficient, such as the center bit and the auger bit. The center bit will cut holes varying from a quarter of an inch to three quarters of an inch in diameter, and is used by pressing the knob end against the chest, and twirling the center part round with the hand.

It cuts a hole very clean, leaving it quite smooth inside. The auger bit is for the same purpose, and is used in the same manner. Another bit, called the taper shell bit, is used for making holes wider, and is a very useful implement.

HOW TO MAKE A WHEELBARROW.

One of the handiest things in a garden is a wheelbarrow, and one of the prettiest for the young carpenter to exercise his ingenuity upon. To make one, take a wide plank or board about an inch and a quarter thick. Proceed to your bench, and having fitted it to its proper position, take your jack plane and plane off the rough, next use your smoothing plane to make it smooth. Then take your pencil and draw upon its side the figure of a wheelbarrow. Then take your compass saw and cut round the marks you have made: to do this you will have to fix your board in the screw of your bench. When this is done take your spoke shave, and shave the edges all round till they are very smooth and even, and you have one side of your barrow. Lay this on another piece of board, and mark the shape of it with your pencil; cut and shave it exactly as you did the first side, so that when finished the two will exactly correspond; then cut a piece off another board for the back and front of the barrow, by the same method you cut the sides, and plane and finish them up in a similar way. Cut some tenons at the end of each exactly to correspond with the mortices on the sides, let them be a trifle larger than the mortices, so that they will drive in tight. Then cut the bottom out neatly, and nail it to the sides. Having proceeded thus far, cut out the legs of your barrow, and nail one on each side. Give each leg a shoulder for the sides to rest upon.

To make the wheel. Take a piece of board, and strike a circle upon it the size you wish your wheel to be of, and with the compass saw cut close round to the stroke; cut out a square hole in the center for the nave to join. Then get the blacksmith to put an iron rim round the wheel to keep it from splitting, and a round pin in each side of the nave, and put a staple in each side of the barrow to keep the wheel in its place. Paint the whole of any color you choose, and you will have a wheelbarrow.

THE WAY TO MAKE A BOX.

First ascertain the size you wish your box to be of. Then cut off your stuff, but take care to cut it a quarter of an inch longer than the size of your box from outside to outside. Should you want it deeper or broader than the length of a deal, the widest of which is generally only eleven inches; suppose, for instance, you wish your box to be 18 inches deep, and you have only 9-inch deal to make it with, you will of course have to join two together, or make what is called in carpentering a *gluejoint*. First, then, after you have cut off your stuff, take your jack plane and “scuffle the rough off,” then put your board edgeways into the bench-screw, and take your trying plane or long plane to get the edge of the deals that are to be glued together perfectly straight and even; and lastly use the joiner plane, which will take off a nice uniform shaving of the whole length of the board. Proceed exactly in the same manner with the other board to be joined to the first. Then, after having made each thoroughly smooth, clap the two together and see if they will lie close in every part; if not you must plane them till they do, taking care to plane the edges perfectly square, or at right angles to the surface of the board, for if you are not careful in this particular, when your boards are glued together they will be of this form. When you have joined them properly for glueing, let your glue be nicely hot and not too thick, and hold both edges of the boards together so that you can with a brush put the glue on both at one time, put the two together very quickly, let one of them be in the bench-screw, and while there rub the other backwards and forwards until the glue sets, which it will soon do if well joined. Let the whole dry, and then the glued part will be as strong as any other part of the board.

After your sides, ends, bottom, and top are thus prepared, you must then plane them up nicely, so that they are perfectly smooth and straight. Use first the jack plane, then the trying plane. When this is done you have to proceed to a nice little job, namely, to dovetail the corners together so as to form your box. In this process much depends upon the planing and squaring of the stuff, for if you have not done this nicely the dovetailing will be very imperfectly performed. Assuming that everything has been well done, then take the two ends of the box, and see that each is perfectly square and true to the other. Then allow one-eighth of an inch more than the thickness of your sides, and set out the ends, squaring it over on both sides, which when the dovetails are cut out will form the inside of the box.

TO CUT THE DOVETAILS.

Take one “end-piece” of the box, and place it endways into the bench-screw, and mark out the dovetails on the edge of the board inside, then with your dovetail saw cut in into the marks down to the lines squared over on the flat side. Then with a chisel cut out that part of the wood that is crossed, and leave the other part, this being the part which will form the pins or tails. Then take one side of your box and lay it flat on the bench, the inside uppermost; then place the end you have cut on it, keeping the edges flush, and mark round the shape of the pins, which will leave their form on the side piece, the black places being the mortices which are to be cut out. In cutting out these you must be careful to cut within side of the stroke, so that the mortices will be a little smaller than the pins, which will admit of their being driven in quite tight, and will allow the glue to adhere to them (for you have to glue these when you fix them). When you have thus put the ends and sides together let them stand till the glue gets dry, then take your planes and plane the quarter of an inch off the pins which you allowed to be a little longer than the length of the box, and you have then made the body of your box.

THE BOTTOM OF THE BOX.

Cut your bottom the exact size of the box, nail the bottom on, and “get out” a piece of wood (by cutting and planing in the usual manner) to nail round so as to form a skirting to it, and at the same time hide the joints of the bottom; “get out” a similar piece of wood to nail round the top which will form the lid. Then get a pair of box joints and a lock, and having put them on by a stroke of your own ingenuity, you will have a “box.”

GALVANISM, OR VOLTAIC ELECTRICITY.

“To play with fire
They say is dangerous; what is it then
To shake hands with the lightning, and to sport
With thunder?”—TYLER.

Galvanism, or electricity of quantity, in contradistinction to frictional electricity, called electricity of intensity, owes its name to the experiments on animal irritability made in 1790 by M. Galvani, a professor of anatomy at Bologna. These experiments were suggested by the following circumstances.

ORIGIN OF GALVANISM.

It happened that the wife of Galvani, who was consumptive, was advised to take as an article of food some soup made of the flesh of frogs. Several of these creatures were killed and skinned, and were lying on the table in the laboratory close to an electrical machine, with which a pupil of the professor was making experiments. While the machine was in action, he chanced to touch the bare nerve of the leg of one of the frogs with the blade of the knife that he had in his hand, when suddenly the whole limb was thrown into violent convulsions. Galvani was not present when this occurred; but being informed of it, he immediately set himself to investigate the cause. He found that it was only when a spark was drawn from the prime conductor, and when the knife or any other good conductor was in contact with the nerve, that the contracting took place; and after a time he discovered that the effect was independent of the electrical machine, and might be equally well produced by making a metallic communication between the outside muscle and the crural nerve.

SIMPLE EXPERIMENT TO EXCITE GALVANIC ACTION.

If the young experimenter will obtain a piece of zinc of the size of half a dollar and place it on the top of his tongue, and place a half-dollar beneath it, and bring the edges of the half-dollar and zinc in contact in front of his

tongue, he will notice a peculiar sensation in the nerves of this organ, and some taste will be imparted to his mouth at the moment of contact.

WITH METAL PLATES IN WATER.

If we take two plates of different kinds of metal, platinum (or copper) and zinc for example, and immerse them in pure water, having wires attached to them above, then if the wire of each is brought into contact in another vessel of water, a galvanic circle will be formed, the water will be slowly decomposed, its oxygen will be fixed on the zinc wire, and at the same time a current of electricity will be transmitted through the liquid to the platina or copper wire, on the end of which the other element of water, namely, the hydrogen, will make its appearance in the form of minute gas bubbles. The electrical current passes back again into the zinc at the points of its contact with the platina, and thus a continued current is kept up, and hence it is called a galvanic circle. The moment the circuit is broken by separating the wires the current ceases, but is again renewed by making them touch either in or out of the water. If a small quantity of sulphuric acid be added to the water, the phenomenon will be more apparent. The end of the wire attached to the piece of platinum or copper is called the positive pole of the battery, and that of the wire attached to the zinc the negative pole.

The current of electricity here generated will be extremely feeble; but this can be easily increased by multiplying the glasses and the number of the pieces of metal. If we take six such glasses instead of one, partially fill them with dilute sulphuric acid, and put a piece of zinc and copper into each, connecting them by means of copper wire from glass to glass through the whole series, a stronger current of electricity will be the result. The experimenter must be careful not to let the wire and zinc touch each other at the bottom of the tumblers, and must also remember that the copper of glass 1 is connected with the zinc of glass 2, and so on.

TO MAKE A MAGNET BY THE VOLTAIC CURRENT.

To effect this, make a connection between the poles of the above or any excited battery with the two ends of a wire formed into a spiral coil, by bending common bonnet-wire closely round a cylinder, or tube, of about an inch in diameter; into this coil introduce a needle or piece of steel wire,

laying it lengthways down the circles of the coil. In a few minutes after the electric fluid has passed through the spiral wire, and consequently round the needle or wire, the latter will be found to be strongly magnetized, and to possess all the properties of a magnet.

EFFECTS OF GALVANISM ON A MAGNET.

If a galvanic current, or any electric current, be made to pass along a wire under which, and in a line with it, a compass is placed, it will be found that the needle will no longer point north and south, but will take a direction nearly across the current, and point almost east and west.

CHANGE OF COLOR BY GALVANISM.

Put a teaspoonful of sulphate of soda into a cup, and dissolve it in hot water; pour a little cabbage blue into the solution, and put a portion into two glasses, connecting them by a piece of linen or cotton cloth previously moistened in the same solution. On putting one of the wires of the galvanic pole into each glass, the acid accumulates in the one, turning the blue to a red, and the alkali in the other, rendering it green. If the wires be now reversed, the acid accumulates eventually in the glass where the alkali appeared, while the alkali passes to the glass where the acid was.

THE GALVANIC SHOCK.

If the ends of the wires of a small galvanic battery are connected with a proper electro-magnetic coil, which may now be purchased at a very cheap rate, and the wires from the coil be placed in separate basins of water, then, on dipping the fingers of each hand in the basin, a smart shock will be felt, with a particular aching accompanied with trembling. With a strong battery and larger coil this effect is felt as high as the shoulders. The shock will also be felt by simply holding the wires of a powerful galvanic battery, one in each hand, provided the hands be moistened with salt and water. Several persons may receive the shock from the battery and coil together by joining hands.

THE ELECTROTYPE.

The electro-galvanic current has in no case been more interestingly employed than in the process of electrotyping. It consists of a mode of obtaining the copy of coins, medals, engraved plates, and other objects, which may be easily illustrated.

HOW TO MAKE AN ELECTROTYPE APPARATUS.

Take an earthen jar and a porous tube; fill the tube with ten parts of water and one of sulphuric acid; put it into the jar, into which pour as much of a solution of sulphate of copper (blue vitriol) as will fill three parts of it; place in the tube a piece of zinc, to which a copper wire is soldered and bent round, so that one end be immersed in the sulphate of copper; and a deposit of the copper will be immediately formed upon the wire. If there be plenty of acid and water, so as to allow of the action enduring for a long time, this process will go on till it has deposited all the copper. This is the principle upon which electrotyping proceeds—a principle referable to electro-chemical decomposition.

TO OBTAIN THE COPY OF A COIN OR MEDAL.

Never place the original medal in the apparatus, or the deposited copper may adhere so tightly to it that the removal destroys the beauty of the medal. Having taken an impression in sealing-wax, cover the latter with black-lead, and attach a wire so that it is in contact with the black-lead. To the wire and cast thus arranged a piece of sheet or cast zinc, amalgamated with mercury, must be attached, and we are at once furnished with the materials for the battery, as the object to be copied supplies the place of the copper. The medal must always be placed horizontally. Now let the apparatus be charged with the solution, by pouring into the outer vessel a portion of the coppery solution, so that it will stand about an inch above the medal; then pour in the glass the dilute acid to the same height as the former; now introduce the zinc into the acid, and the object to be copied into the solution of copper, which will immediately be deposited on the medal, and when of a sufficient thickness may be taken off.

HEAT.

HEAT, OR CALORIC.

The chief agent in causing the repulsion or separation of the particles of bodies from each other is heat, or more correctly *caloric*, by which is understood the unknown *cause* of the effect called heat. Philosophers are not agreed upon the nature of this wonderful agent. It pervades all nature, is the cause of nearly all the changes that take place both in organic and inorganic matter, and has great influence in the meteorological phenomena which we observe in the atmosphere that surrounds our planet. It appears to be intimately connected with light, electricity, and magnetism—subjects which the genius of Faraday and others have investigated, and by their discoveries brought us nearer to the knowledge of the real nature of these most wonderful forces.

Caloric, then, exists in all bodies, and has a constant tendency to equalize itself, as far at least as its outward manifestation, called temperature, is concerned; for if a *hot* body be brought near colder ones, it will give up heat to them, until by its loss and their gain they all become of the same temperature; and this proceeds more or less rapidly, according as the original difference of temperature was greater or less. Some other circumstances also influence this equalization. The converse will take place on introducing a cold body among warmer ones, when heat will be abstracted from all the bodies within reach of its influence, until it has absorbed sufficient caloric to bring its own temperature to an equality with theirs. This is the true explanation of the apparent production of *cold*. When, for instance, an iceberg comes across a ship's course, it appears to *give out* cold, whereas it has abstracted the heat from the air and sea in its neighborhood, and they in turn act upon the ship and everything in it, until one common temperature is produced in all the neighboring bodies.

It does not follow that the bodies thus equalized in temperature contain equal quantities of caloric; far from it. Each body requires a particular quantity of caloric to raise its temperature through a certain number of degrees; and such quantity is called its *specific* caloric. A pound of water, for instance, will take just twice as much caloric as a pound of olive oil, to

raise its temperature through the same number of degrees; the *specific caloric* of water is therefore double that of oil. Mix any quantity of oil at 60 deg. of temperature with an equal weight of water at 90 deg., and you will find the temperature of the mixture to be nearly 80 deg., instead of only 74 deg. or 75 deg., showing that while the water has lost only 10 deg. of caloric, the mixture has risen 20 deg. If the oil be at 90 deg., and the water at 60 deg., the resulting temperature will be only 70 deg., or thereabouts, instead of 75 deg., the mean; thus, here the hot oil has lost 20 deg., while the mixture has risen only 10 deg.; the water, then, contains at the same temperature *twice* as much caloric as the oil; its specific caloric is *double* that of the oil. This mean temperature does result when equal weights of the same body at different temperatures are mixed together.

The sensations called heat and cold are by no means accurate measures of the real temperature of any substances, for many causes influence these sensations, some belonging to the substances themselves, others to the state of our organs at the time. Every one has remarked that metals in a warm room feel warmer, and in a cold room colder than wooden articles, and these again than woollen or cotton articles of dress or furniture; this arises from metals being what is termed better *conductors* of heat than wood, and this better than wool, &c., that is, they give out or absorb caloric more rapidly than these last. Some philosophers, wishing to ascertain how much heat the human body could endure, had a room heated with stoves, every crevice being carefully stopped, until the temperature rose so high that a beefsteak placed on the table was sufficiently cooked to be eaten. They were dressed in flannel, and could with impunity touch the carpets, curtains, &c., in the room; but the iron handles, fire-irons, and all metallic substances, burnt their fingers; and one who wore silver spectacles was obliged to remove them to save his nose. The fallacy of our sensations may be easily shown by taking two basins, placing in one some water at 100 deg., in another some water at as low a temperature as can easily be procured—hold the right hand in one, the left in the other, for a few minutes, and then mix them, and place both hands in the mixture; it will feel quite *cold* to the hand that had been in the hotter water, and *hot* to the other.

In order to arrive at a correct estimate of the temperature of bodies, instruments are made use of called thermometers, or measurers of heat,

which show increase or diminution of temperature by the rising or falling of a column of some fluid in a tube of glass, one end of which is expanded into a bulb, and the other hermetically sealed. This effect is produced by the expansion or swelling of the fluid as caloric is added to, and its contraction when caloric is abstracted from it. Colored spirits of wine, or quicksilver, are the most usual thermometric fluids, and the tube containing them is fixed to a wooden or metallic frame, on which certain divisions are marked, called degrees.

That in general use in America is called Fahrenheit's from the name of the person who first introduced that particular scale. In this thermometer, the point at which the mercury in the tube stands when plunged into melting ice, is marked 32 degrees, and the distance between that point and the point to which the mercury rises in boiling water is divided into 180 equal parts, called degrees; so that water is said to boil at 212 degrees = 180 degrees + 32 degrees. There are two other scales of temperature used in different parts of the world, but it is not worth while to notice them here.

Not only do different bodies at the same degree of temperature contain very different quantities of caloric, but this also is the case with the same body in different forms. Ice, water, and steam are three forms of the same body, but ice at 32 degrees contains much less caloric than water at the same temperature, and water at 212 degrees contains much less caloric than steam (or water in a state of vapor) at that temperature.

Place in a jar any given quantity of snow, or small pieces of ice, at 32 degrees, and in another the same weight of water at 32 degrees, pour on each an equal weight of water at 172 degrees, and you will find that in the first case the ice will be melted, but the temperature will remain at 32 degrees, or thereabouts, while the temperature of the water in the other vessel will have risen to 100 degrees or thereabouts, being as near as possible the half of the excess of the temperature of the hot water, 140 degrees over that of the cold, namely 70 degrees added to 32 degrees, the original temperature. Now, what has become of the heat which was added to the ice, and is apparently lost?—it is *absorbed* by the ice in its passage to the fluid state; so that water may be said to be a compound of ice and caloric.

Again, take 10 ounces of water at about 50 degrees, and add 1 oz. of water at 212 degrees, and the temperature of the mixture will be about 66 degrees; then condense some steam at 212 degrees into another 10 oz. of water until it has become 11 oz., and you will find the temperature will be nearly 212 degrees. Why does the ounce of steam at 212 degrees raise the temperature of the water so much higher than the ounce of water at the same temperature? Obviously because it contains hidden in its substance a vast quantity of caloric, not to be detected by the thermometer; in fact, that steam is a compound of *water* and caloric, as water is a compound of *ice* and caloric; and this caloric which exists, more or less, in all bodies without producing any obvious effect, is called *latent* caloric, from the Latin verb *lateo*, to lie hid. The quantity of caloric thus absorbed, as it were, by various bodies, differs for each body, and for the same body in different forms, as mentioned above.

EXPANSION.

As a general rule, all bodies, whether solid, liquid, or gaseous, are expanded by caloric. This may be shown by experiments in each form of matter.

Have a small iron rod made, which when cold just passes through a hole in a plate of metal; heat it, and it will no longer pass; after a time the rod will return to its former temperature, and then will go through the hole as before. The rod increases in length as well as width; if you have a gauge divided into $\frac{1}{100}$ of an inch, and place the rod in it when cold, noting its position, on heating it will extend to a greater length in the gauge, returning to its former place when cool.

The effect of caloric in causing fluids to expand is actually employed as a measure of quantity in the thermometer, the rise of the fluid in the tube when heated depending on the increased bulk of the fluid occasioned by the addition of caloric. The same fact is to be noticed every day when the cook fills the kettle, and places it on the fire. As the water becomes warmer it expands, that is, takes up more room than it did before, and the water escapes by slow degrees, increasing as the heat increases, up to the point of boiling, when a sudden commotion takes place from the condensation of a portion of the water into steam.

But it is in the form of vapor or gas (which, by the bye, is not the same thing), that the expansive force of caloric is most obvious. The gigantic powers of the steam-engine depend entirely on the tendency of vapor to expand on the addition of caloric; and this force of expansion appears to have no limit; boilers made of iron plates an inch or even more in thickness, and the buildings or ships containing them, having been torn to pieces and scattered in all directions by the expansive power of steam. Take a bladder and fill it about half-full of air, and tie the neck securely; upon holding it to the fire it will swell out and become quite tense from the expansion of the contained air.

The principal source of caloric is the sun, whose beams, diffused through all nature by the refractive property of the atmosphere, are the source of vitality both to vegetables and animals, and when concentrated by a large convex lens, produce the most intense heat, sufficient to light a piece of diamond, and melt platinum. Caloric is also produced or evolved by combustion, by friction, percussion, chemical combination, electricity, and galvanism.

The evolution of heat by friction may be witnessed daily in a thousand instances. Lucifer matches are lighted by rubbing the highly inflammable substances with which they are tipped against a piece of sand-paper. Nearly all savage people procure fire by rubbing a piece of hard wood violently against a softer piece. The axle-trees of steam-engines, and even of carriages, have been known to be so heated by friction as to endanger burning the carriage; and it is very usual to be obliged to pour a quantity of cold water on the iron axle of the carriages of an express train after an hour of constant and rapid work. If you merely rub the blade of a knife rapidly on a piece of wood it will become hot enough to burn your hand.

Percussion is merely a more energetic kind of friction, and is often resorted to by the blacksmith to light his furnace. He places a nail or other piece of soft iron on his anvil, and beats it rapidly with the hammer, when it becomes actually red hot. The production of sparks by striking flint against steel, or two pieces of flint one against the other, are familiar instances of heat produced by percussion.

One of the most powerful means of producing heat is the process of combustion.

Combustion, as the word imports, is the *burning together* of two or more substances, a chemical union of oxygen generally with carbon and hydrogen in some shape or other. In our ordinary fires we burn coal, a hydro-carbon as it is called; and the gas which is now so universally used for the purpose of illumination, is a compound of the same bodies—so wax, tallow, oil of various kinds, both of animal and vegetable origin, are all hydro-carbons.

On the application of a sufficient heat, and a free access of atmospheric air, or of some other gas containing oxygen in a certain state of combination, these bodies take fire, and continue to burn either with flame, or a red or even white heat without flame, until they are consumed; that is, until they have entered into new combinations with the oxygen, and are converted into carbonic acid and water, the carbon forming the first product, the hydrogen the other.

The following experiment shows the productions of heat by chemical action alone. Bruise some fresh prepared crystals of nitrate of copper, spread them over a piece of tin foil, sprinkle them with a little water; then fold up the foil tightly as rapidly as possible, and in a minute or two it will become red-hot, the tin apparently burning away. The heat is produced by the energetic action of the tin on the nitrate of copper, taking away its oxygen in order to unite with the nitrate acid, for which, as well as for the oxygen, the tin has a much greater affinity than the copper has.

Combustion without flame may be shown in a very elegant and agreeable manner, by making a coil of platinum wire by twisting it round the stem of a tobacco-pipe, or any cylindrical body, for a dozen times or so, leaving about an inch straight, which should be inserted into the wick of a spirit-lamp; light the lamp, and after it has burnt for a minute or two extinguish the flame quickly; the wire will soon become red-hot, and, if kept from draughts of air, will continue to burn until all the spirit is consumed. Spongy platinum, as it is called, answers rather better than wire, and has been employed in the formation of fumigators for the drawing-room, in which, instead of pure spirits, some perfume, such as lavender water, is used; by its combustion an agreeable odor is diffused through the apartment. These little lamps were much in vogue a few years ago, but are now nearly out of fashion.

Experiments on combustion might be multiplied almost to any amount, but the above will be sufficient for the present. When we come to treat of the properties of the gases and some other substances, we shall have occasion to recur to this subject.

The production of caloric by chemical combination may be exhibited by mixing carefully one part of oil of vitriol with two of water, when sufficient heat will be produced to boil some water in a thin and narrow tube, which may be used as a rod to stir the mixture.

The production of heat by electric and galvanic agency belongs to another subject.

HYDRAULICS.

The science of Hydraulics comprehends the laws which regulate non-elastic fluids in motion, and especially water, etc.

Water can only be set in motion by two causes—the pressure of the atmosphere, or its own gravity. The principal law concerning fluids is, that they always preserve their own level. Hence water can be distributed over a town from any reservoir that is higher than the houses to be supplied; and the same principle will enable us to form fountains in a garden, or other place. Should any of our young readers wish to form a fountain, they may, by bringing a pipe from a water-tank, which should be at the upper part of the house, convey the water down to the garden. Then, by leading it through the earth, underneath the path or grass-plot, and turning it to a perpendicular position, the water will spring out, and rise nearly as high as the level of that in the tank. The pipe should have a faucet, so that the water may be let on or shut off at pleasure.

THE SYPHON.

The syphon is a bent tube, having one leg shorter than the other. It acts by the pressure of the atmosphere. In order to make a syphon act, it is necessary first to fill both legs quite full of the fluid, and then the shorter leg must be placed in the vessel to be emptied. Immediately upon withdrawing the finger from the longer leg, the liquor will flow.

THE PUMP.

The action of the common pump is as follows: When the handle is raised, the piston-rod descends, and brings the piston-valve—called the sucker, or bucket—to another valve, which is fixed, and opens inward towards the piston. When the handle is drawn down, the piston is raised, and, as it is air-tight, a vacuum is produced between the two valves; the air in the barrel of the pump, betwixt the lower valve and the water, then forces open the lower valve, and rushes through to fill up this vacuum; and the air in the pump being less dense than the external atmosphere, the water is forced a short way up the barrel. When the piston again descends to the lower valve, the

air between them is again forced out by forcing open the upper valve; and when the piston is raised, a vacuum is again produced, and the air below the lower valve rushes up, and the water in consequence is again raised a little further. This operation continues until the water rises above the lower valve; at every stroke afterwards, the water passes through the valve of the descending piston, and is raised by it, on its ascent, until it issues out of the spout.

THE HYDRAULIC DANCER.

Make a little figure of cork, in the shape of a dancing mountebank, sailor, etc. In this figure place a small hollow cone, made of thin leaf brass. When this figure is placed upon any jet, such as that of the fountain recommended to be constructed, it will be suspended on the top of the water, and perform a great variety of amusing motions. If a hollow ball of very thin copper, of an inch in diameter, be placed on a similar jet, it will remain suspended, turning round and spreading the water all about it.

MAGNETISM.

The attractive power of the loadstone has been known from a very remote period. The natural magnet appears native in a gray iron ore in octahedral crystals, composed of 168 parts of iron, and 64 parts of oxygen. Its properties seem to have been studied in Europe during the dark ages and a directive power is alluded to by Cardinal James de Vitri, who flourished about the year 1200, who observed that it was indispensable to those who travel much by sea.

In modern times, the history as well as the nature of the magnet has engaged remarkable attention; and it has been determined beyond all dispute that the magnet was used by the Chinese under the name of the *tche-chy* (directing stone) about 2604 years before Christ. It passed from them to the Arabs, and was first used in Europe after the crusades; and Ludi Vestomanus asserts that about the year 1500 he saw a pilot in the East Indies direct his course by a magnetic needle like those now in use.

TO MAKE ARTIFICIAL MAGNETS.

This may be done by stroking a piece of hard steel with a natural or artificial magnet. Take a common sewing needle and pass the north pole of a magnet from the eye to the point, pressing it gently in so doing. After reaching the end of the needle the magnet must not be passed back again towards the eye, but must be lifted up and applied again to that end, the friction being always in the same direction. After repeating this for a few times the needle will become magnetized, and attract iron filings, etc.

HOW TO MAGNETIZE A POKER.

Hold it in the left hand in a position slightly inclined from the perpendicular, the lower end pointing to the north, and then strike it smartly several times with a large iron hammer, and it will be found to possess the powers of a magnet, although but slightly.

TO SHOW MAGNETIC REPULSION AND ATTRACTION.

Suspend two short pieces of iron wire, so that they will hang in contact in a vertical position. If the north pole of a magnet be now brought to a moderate distance between the wires, they will recede from each other.

The ends being made south poles by induction from the north pole, will repel each other, and so will the north poles. This separation of the wires will increase as the magnet approaches them, but there will be a particular distance at which the attractive force overcomes the repulsive force of the poles, and causes the wires to converge.

NORTH AND SOUTH POLES OF THE MAGNET.

Each magnet has its poles, north and south—the north or south poles of one magnet repel the north and south pole of another. If a magnet be dipped in some iron filings, they will be immediately attracted to one end. Supposing this to be the north pole, each of the ends of the filings, not in contact with the magnet, will become north poles, while the ends in contact will by induction become south poles. Both will have a tendency to repel each other, and the filings will stand on the magnet.

POLARITY OF THE MAGNET.

The best method of proving this is to take a magnet or a piece of steel rendered magnetic, and to place it on a piece of cork by laying it in a groove cut to receive it. If the cork be placed in the center of a basin of water, and allowed to swim freely on its surface, so that it is not attracted by the sides of the basin, it will be found to turn its north pole to the north, and its south pole to the south, the same as the mariner's compass. If you fix two magnets in two pieces of cork, and place them also in a basin of water, and they are in a parallel position with the same poles together, that is, north to north, and south to south, they will mutually repel each other; but if the contrary poles point to one another, as north to south, they will be attracted.

THE MAGNETIC FISH.

Fish are to be purchased at the toy-stores, by which the young “magnétique” may perform this experiment; they are made hollow, and will float on the water. In the mouth of each should be inserted a piece of magnetic wire. The angling rod is like any other rod, and has a silken thread

for a line, and an iron hook also strongly magnetized. To catch the fish it is only necessary to put the hook in contact with the noses of the fish, and they will be taken without any bait.

THE MAGNETIC SWAN.

The figure of a swan should be cut in cork, and within its beak a small strongly magnetized piece of steel should be placed. The swan should then be covered with a coating of white wax, and fashioned further into the shape of a swan, and glass beads may be placed in its head for eyes. This should be placed in a small tub or large basin of water, and to make it swim about, you should place in a white stick about nine inches long a magnetic bar, on which the north and south poles are marked. If you wish to bring the swan towards you, present to him the north pole of the wand, if you wish it to retire, present the south pole, and thus you may direct the swan to any part you desire.

TO SUSPEND A NEEDLE IN THE AIR BY MAGNETISM.

Place a magnet on a stand to raise it a little above the table; then bring a small sewing needle containing a thread, within a little of the magnet, keeping hold of the thread to prevent the needle from attaching itself to the magnet. The needle in endeavoring to fly to the magnet, and being prevented by the thread, will remain curiously suspended in the air, reminding us of the fable of Mahomed's coffin.

TO MAKE ARTIFICIAL MAGNETS WITHOUT THE AID EITHER OF NATURAL LOADSTONES OR ARTIFICIAL MAGNETS.

Take an iron poker and tongs, or two bars of iron, the larger and the older the better, and fixing the poker upright, hold to it with the left hand near the top by a silk thread, a bar of soft steel about three inches long, one-fourth of an inch broad and one-twentieth thick; mark one end, and let this end be downwards. Then grasping the tongs with the right hand a little below the middle, and keeping them nearly in a vertical line, let the bar be rubbed with the lower end of the tongs, from the marked end of the bar to its upper end about ten times of each side of it. By this means the bar will receive as

much magnetism as will enable it to lift a small key at the marked end; and this end of the bar being suspended by its middle, or made to rest on a joint, will turn to the north, and is called its north pole, the unmarked end being the south pole. This is the method recommended by Mr. Caxton, in his process, which he regarded superior to those in former use, and of which a more detailed account will be found in his interesting volume.

HORSE-SHOE MAGNETS.

The form of a horse-shoe is generally given to magnetized bars, when both poles are wanted to act together, which frequently happens in various experiments, such as for lifting weights by the force of magnetic attraction, and for magnetizing steel bars by the process of double touch, for which they are exceedingly convenient. The following is the method of making a powerful magnetic battery of the horse-shoe form. Twelve bars or plates of steel are to be taken, and having been previously bent to the required form, that is, the horse-shoe shape, they are then bound together by means of rivets at their ends; before being finally fastened they are each separately magnetized and afterwards finally united.

Horse-shoe magnets should have a short bar of soft iron adapted to connect the two poles, and should never be laid by without such a piece of iron adhering to them. Bar magnets should be kept in pairs with their poles turned in contrary directions, and they should be kept from rust. Both kinds of magnets have their power not only preserved but increased, by keeping them surrounded with a mass of dry filings of soft iron, each particle of which will re-act by its induced magnetism upon the point of the magnet to which it adheres, and maintain in that point its primitive magnetic state.

EXPERIMENT TO SHOW THAT SOFT IRON POSSESSES MAGNETIC PROPERTIES WHILE IT REMAINS IN THE VICINITY OF A MAGNET.

Let a magnet and a key be held horizontally near one of its poles, or near its lower edge. Then if another piece of iron, such as a small nail, be applied to the other end of the key, the nail will hang from the key, and will continue to do so while the magnet is slowly withdrawn; but when it has been removed beyond a certain distance, the nail will drop from the key, because

the magnetism induced in the key becomes at that distance too weak to support the weight of the nail. That this is the real cause of its falling off may be proved by taking a still lighter fragment of iron, such as a piece of very slender wire, and applying it to the key. The magnetism of the key will still be sufficiently strong to support the wire, though it cannot the nail, and it will continue to support it even when the magnet is yet further removed; at length, however, it drops off.

ELECTRO-MAGNETISM.

The identity of magnetism with electricity alluded to in a former paragraph, has led to the formation of a new science under the above name, and to some of the interesting experiments connected with it, we shall briefly allude for the amusement of the young reader.

POWER OF THE ELECTRO-MAGNET.

The same influence which affects the magnetic needle already described, will also communicate magnetism to soft iron. If a bar of that metal bent, be surrounded with a common bonnet wire, or a copper wire prevented from touching the iron by a winding of cotton or thread, and then if a current of voltaic electricity be sent through the wire, the bar becomes a powerful magnet, and will continue so as long as the connection with the battery is preserved. On breaking the contact, the magnetism disappears. This experiment may be easily made by the young reader with a horse-shoe magnet, surrounded by several coils of wire.

THE MARINER'S COMPASS AND EXPERIMENTS WITH A POCKET COMPASS.

The mariner's compass is an artificial magnet fitted in a proper box, and consists of three parts—the box, the card or fly, and the needle. The box is suspended in a square wooden case, by means of two concentric brass circles called gimbals, so fixed by brazen axes to the two boxes, that the inner one, or compass-box, retains a horizontal position in all motions of the ship. The card is a circular piece of paper which is fastened upon the needle, and moves with it. The outer edge of the card is divided into thirty-two points, called points of the compass. The needle is a slender bar of

hardened steel, having a hollow agate cup in the center, which moves upon the point of a pivot made of brass.

VARIATION OF THE NEEDLE.

The magnetic needle does not point exactly north and south, but the north pole of the needle takes a direction to the west of the true north. It is constantly changing, and varies at different parts of the earth, and at different times of the day.

DIP OF THE NEEDLE.

Another remarkable and evident manifestation of the influence of the magnetism of the earth upon the needle is the inclination or dip of the latter which is a deviation from its horizontal place in a downward direction in northern regions of its north, and in southern regions of its south pole. In balancing the needle on the card, on account of this dipping, a small weight or movable piece of brass is placed on one end of the needle, by the shifting of which either nearer to or further from the center, the needle will always be balanced.

USEFUL AMUSEMENT WITH THE POCKET COMPASS.

Pocket compasses are to be bought for from 50 cents to \$1, and may be used in many ways. In traveling over mountains or a wide extended plain, they are indispensably necessary, and no one should go on a tour without such a companion; it will be a very useful and amusing exercise for any young person to take the bearings of his own or some particular locality, and make out what may be called a bearing card. This he may easily do in the following manner: Supposing he wishes, for instance, to take the bearings of his own house, he has nothing to do but set his pocket compass upon a map of the district,—a county map will do very well, unless his house stands on the verge of a county, then two county maps will be necessary. He must make the north of the map exactly coincide with the north, as indicated by his compass, and having fixed his map in this situation, he should take a ruler and piece of paper, and dot down the exact bearings of each important town, or place, or village, around him. Let him suppose himself, for instance, in the town of Albany, N. Y., and laying down

his map as indicated by the compass, north to north and south to south, he will find the following places due north, Balston Spa; Hudson, south; Schoharie, west. The other points of the compass may be filled up in the same manner. Should, therefore, our young friend be upon any other elevated situation near his own dwelling, or upon any other elevated spot from which the bearings have been taken, he will be able to inform his young friends that such and such a place lies in such a direction, that this place lies due north, the other north-west, a third south-east, the fourth south-west, etc., etc.

INTERESTING PARTICULARS CONCERNING THE MAGNET.

Fire-irons which have rested in an upright position in a room during the summer months are often highly magnetic.

Iron bars standing erect, such as the gratings of a prison cell, or the iron railings before houses, are often magnetic.

Great iron-clad ships are powerfully magnetic, and therefore affect the compass by which the vessel is steered; ingenious arrangements are therefore made to correct the effect of the local attraction, so that the man-of-war may be steered correctly.

Magnetism may be made to pass through a deal board; to exhibit which, lay a needle on the smooth part above, and run a magnet along the under side, and the needle will be found to follow the course of the magnet. A magnet dipped into boiling water loses part of its magnetism, which, however, returns upon its cooling.

A sudden blow given to a magnet often destroys its magnetic power.

HOW TO BECOME A PHOTOGRAPHER.

Associated with the use of iodine and bromine is an art which every intelligent boy may practice, if he will attend to the following precise details kindly furnished by an experienced photographer.

HOW TO MAKE THE NEGATIVE ON GLASS, USING COLLODION BROMIODIZED FOR IRON DEVELOPMENT.

1. The edges of the glass should be ground all round, also slightly on the surface of the edges. This prevents contraction of the film, enabling it to resist the action of a heavy stream of water. Mark one side in the corner with a diamond, and upon this side bestow the greatest care.

2. *To clean the glass, if new.*—Make a mixture of spirits of wine and solution of ammonia, equal parts; render it as thick as cream with tripoli; with a piece of cotton-wool kept for this purpose rub a small quantity over that side marked as described, wash well under a tap of water, and wipe dry with a piece of old linen, washed without soap, and kept scrupulously clean for this purpose. Plates should not, however, be cleaned in the operating room with the above mixture; the vapor of ammonia might prove injurious to the chemicals.

3. Now polish with an old white silk handkerchief. If this latter precaution be not taken, small particles of linen will be left upon the plate: these are perhaps only seen when draining off the collodion; they form nuclei and eddies, checking the collodion in its course. Some of these minute fibers are washed off, and contaminate the next picture. To all lovers of clean pictures our advice therefore is, having well dried the plate with old linen, lay it, clean side upwards, upon a few sheets of common glazed demy paper (not blotting), and rub it hard with the silk until sensibly warm; this has the double advantage of dispersing fibers and moisture, for all glass plates are slightly in a hygrometric condition. Double the silk rubber up to form a pad, and with this the glass must be firmly dusted down just before pouring on the collodion, which will then run most evenly; if the coated plate is now viewed by transmitted light, not a speck or blemish will be seen upon it.

When a plate cleaned as above described is breathed upon, the moisture does not evaporate slowly, but *flies off*. Do not be afraid of putting the glass into an electrical condition with the silk rubber; on this account objections have been raised to the use of silk; practically, however, I find it a most valuable auxiliary in this starting-point of the process, the perfect manipulation of which makes an important difference in the value of the finished picture. What can be more inartistic and annoying to an educated eye than spots, patches, stars, and sky-rockets, the forms and shapes of which rival, in numberless variety, a display of fireworks? Let us not, therefore, be contented with pictures, however good in other respects, presenting these deformities—so many blots on the photographic escutcheon.

To clean a glass after having used it, when not varnished.—Wash off the collodion film with water, then clean the marked side with plain tripoli and water, and dry as above.

To coat the plate.—First remove all the particles of dried collodion from the mouth of the bottle. Now pour upon the center of the cleaned glass as much collodion as it will hold. Do not perform this operation hurriedly, take time, and systematically incline the plate in such a manner that the collodion may run into each corner in succession; when perfectly covered, pour off gently the excess into the bottle at one of the corners nearest to you; with observation and practice dexterity is easily acquired. There are many ways of coating the plate; each person will adopt that which practice teaches him is best. The pneumatic plate-holder is a convenient little instrument to use for holding the plate whilst pouring on the collodion; it may be used for both small and large plates.

Keep the corner of the glass plate in contact with the neck of the bottle whilst pouring off the collodion; otherwise the film will be wavy in places.

4. As soon as the collodion ceases to run, plunge the prepared glass gently, without stopping, into the nitrate of silver bath, which is prepared as follows: Into a 20-oz. stoppered bottle put nitrate of silver, 1 $\frac{1}{4}$ ozs.; distilled water, 4 ozs.; dissolve. To this solution add iodide of potassium, 4 grs., dissolved in one drachm of distilled water. Mix these two solutions; the precipitate (iodide of silver) thus formed is by shaking entirely dissolved. Add 16 ozs. of distilled water, when the excess of iodide of silver is again

thrown down, but in such a finely divided state as to render the saturation of the bath with iodide of silver perfect. Now drop in sufficient of the oxide of silver to turn the turbid yellow solution a dirty brown color; so long as this effect is produced the quantity of oxide of silver, however much in excess, is of no consequence; shake the bottle well for ten minutes or so at intervals; then add alcohol, 30 minims, and filter; to the filtered solution add dilute nitric acid of the strength stated, 5 minims. The bath is now ready for use, and should be quite neutral.

5. Allow the prepared glass to remain in this bath from five to ten minutes, according to the temperature. Move it up and down three or four times whilst in the bath, in order to get rid of the greasy appearance on the surface; drain it, but not too closely. When in the frame, place upon the back a piece of common blotting-paper, to absorb moisture, and the two lower silver wires should also be covered with slips of blotting-paper; after which the sooner it is placed in the camera the better.

6. The time of exposure can only be ascertained by practice—no rules can be laid down; and I am unacquainted with any royal road, but that of experience, leading to constant success in this most important point.

7. The plate having been taken from the camera and placed upon a leveled stand, or held in the hand, develop immediately the latent image with the following solution:

Iron developing solution.—Protosulphate of iron, $\frac{1}{4}$ oz.; glacial acetic acid, $\frac{1}{4}$ oz.; spirits of wine, $\frac{1}{2}$ oz.; distilled water, 8 ozs.; mix. Pour on of this solution only enough to cover the plate easily, commencing at that edge of the negative which stood uppermost in the camera; move the solution to and fro until it has become intimately mixed with the silver on the plate; then pour off into the developing glass, and *at once* return it on to the plate. When as much intensity has been obtained as possible with the iron developer, it should be thoroughly removed by washing with water. Any intensity may be obtained afterwards by using either of the following solutions:

8. *Intensifying solution.*—Pyrogallic acid, 6 grs.; glacial acetic acid, $\frac{1}{4}$ oz.; distilled water, 6 ozs.; mix. A few drops of a 30-gr. solution of nitrate of silver, the quantity to be regulated according to the intensity required, to be

added, at the moment of using, to as much of the pyrogallic solution as may be necessary.

Intensifying solution (another form).—1. Pyrogallic acid, 8 grs.; citric acid, 20 grs.; distilled water, 2 ozs. 2. Nitrate of silver, 8 grs.; distilled water, 2 ozs. Mix small quantities of the solutions 1 and 2, in equal portions, the moment before using.

The pyrogallic solution, made with good acetic acid, may be kept for a month or more in a cool place. Nevertheless, if the conditions of light and situation are unfavorable, I should prefer this solution just made. The iron solutions act best when freshly prepared.

It is supposed by some that a prolonged action of the iron developer produces fogginess. This may be the case when impure or improperly prepared collodion is used, but certainly not when the preparation is pure and of the proper quality.

When the image is sufficiently intense, wash freely with common filtered water; then pour on a saturated solution of hyposulphate of soda, which should immediately remove the iodide of silver: wash again well with water; allow as much as the plate will hold to soak in for at least a quarter of an hour, changing the water occasionally, to remove all traces of hyposulphate; lastly, wash the plate with a little distilled water, stand up to dry, and, if required, varnish either with spirit or amber varnish.

The following solution is also very commonly used for fixing the negative:—Cyanide of potassium, $\frac{1}{4}$ oz.; water, 12 ozs.

Attention to the following rules and cautions will assist the operator in the production of perfect pictures:—

1. Do not disturb the deposit which will occasionally be found at the bottom of the bottle containing the collodion.
2. Remove all particles of dried film from the neck of the bottle before pouring the collodion on the plate.
3. Never use damp cloths, leathers, or buffs, for giving the final polish to the plate. Negatives with an indistinct and muddy surface are frequently produced from this cause.

4. Let the film set properly before immersion in the nitrate of silver bath: its condition can be ascertained by gently touching the lower part of the coated plate with the end of the finger.
5. Never omit to pass a broad camel-hair brush over the plate just before pouring on the collodion.
6. Bear in mind that, as light is the producing agent, so will it prove a destructive one: not less than four folds of yellow calico should be used to obstruct white light; and in that case the aperture covered should be no larger than is necessary to admit sufficient light for working by. Examine occasionally the yellow calico: when this material is used to exclude white light, it becomes bleached by constant exposure. Do not trust alone to any colored glass; no glass yet made is anti-actinic under all aspects of light and conditions of exposure.
7. When the negative requires intensifying, carefully wash off all traces of the first developing solution before proceeding to intensify. This operation may be performed either before or after the iodide is removed by fixing.
8. Glass baths are preferable to porcelain, ebonite, or gutta-percha baths for solution of nitrate of silver.
9. In using either spirit or amber varnish, before pouring it off, keep the plate horizontal a few seconds. This gives time for soaking in, and prevents the formation of a dull surface arising from too thin a coating.
10. Rub the lenses occasionally with a soft and clean wash-leather, the rapidity of action is much influenced by the brightness of the lenses: their surfaces are constantly affected by moisture in the atmosphere, which condensing, destroys the brilliancy of the image.
11. The white blotting-paper used for some photographic purposes is not suitable for filtering solutions; that only should be employed which is made for this purpose, and is sold under the name of filtering-paper.
12. *Hyposulphate of soda*.—A great deal of rubbish is sold under the name of this salt. As a test of its quality, $1\frac{1}{2}$ drachms should entirely dissolve in 1 drachm of water, and this solution should dissolve rather more than $4\frac{1}{2}$ grains of iodide of silver.

13. *Chemicals*.—The purity of photographic chemicals cannot be too strongly urged; the cheapest are not always the most economical. The commercial preparations are generally not to be depended upon, as these, though perhaps unadulterated, are, strictly speaking, not chemically pure. It is best to procure them from well-known chemists, who understand the purpose for which they are intended, and make the preparation of these substances peculiarly a branch of their business.

14. Never leave chemical solutions exposed in dishes: when done with, pour them back into glass-stoppered bottles, and decant for use from any deposit, or filter if necessary.

15. In all photographic processes it is absolutely necessary to be chemically clean; and this sometimes is not easy. As a rule, never be satisfied with cleanly appearances only, but take such measures as shall insure the absence of all extraneous matter in preparing the solutions, cleaning the glasses, dishes, etc.

16. All stains on the hands, linen, etc., may be removed by means of cyanogen soap or cyanide of potassium, which should be applied without water at first, then thoroughly washed off. To assist the operation, the hands may be now gently rubbed with a fine piece of pumice-stone, when the stains quickly disappear.

For more perfect and complete directions, the reader is referred to any complete work on photography.

MECHANICS.

There is no subject of such importance as Mechanics, as its principles are founded upon the properties of matter and the laws of motion; and in knowing something of these, the tyro will lay the foundation of all substantial knowledge.

The properties of matter are the following: Solidity (or Impenetrability), Divisibility, Mobility, Elasticity, Brittleness, Malleability, Ductility, and Tenacity.

The laws of motion are as follows:—

1. Every body continues in a state of rest or of uniform rectilineal motion, unless affected by some extraneous force.
2. The change of motion is always proportionate to the impelling force.
3. Action and reaction are always equal and contrary.

EXPERIMENT OF THE LAW OF MOTION.

In shooting at “taw,” if the marble be struck “plump,” as it is called, it moves forward exactly in the same line of direction; but if struck sideways, it will move in an oblique direction, and its course will be in a line situated between the direction of its former motion and that of the force impressed. This is called the resolution of forces.

BALANCING.

The center of gravity in a body is that part about which all the other parts equally balance each other. In balancing a stick upon the finger, or upon the chin, it is necessary only to keep the chin or finger exactly under the point which is called the center of gravity.

THE PRANCING HORSE.

Cut out the figure of a horse, and having fixed a curved iron wire to the under part of its body, place a small ball of lead upon it. Place the hind legs of the horse on the table, and it will rock to and fro. If the ball be removed,

the horse would immediately tumble, because unsupported, the center of gravity being in the front of the prop; but upon the ball being replaced, the center of gravity immediately changes as position, and is brought under the prop, and the horse is again in equilibrio.

**TO CONSTRUCT A FIGURE, WHICH BEING PLACED
UPON A CURVED SURFACE, AND INCLINED IN ANY
POSITION, SHALL, WHEN LEFT TO ITSELF, RETURN TO
ITS FORMER POSITION.**

The feet of the figure rest on a curved pivot, which is sustained by two loaded balls below; for the weight of these balls being much greater than that of the figure, their effect is to bring the center of gravity of the whole beneath the point on which it rests; consequently the equilibrium will resist any slight force to disturb it.

**TO MAKE A CARRIAGE RUN IN AN INVERTED
POSITION WITHOUT FALLING.**

It is pretty well known to most boys, that if a tumbler of water be placed within a broad wooden hoop, the whole may be whirled round without falling, owing to the centrifugal force. On the same principle, if a small carriage be placed on an iron band or rail, it will ascend the curve, become inverted, and descend again, without falling.

**TO CAUSE A CYLINDER TO ROLL BY ITS OWN WEIGHT
UP-HILL.**

Procure a coffee-canister, and loading it with a piece of lead, which may be fixed in with solder, the position of the center of gravity is thus altered. If a cylinder so constructed be placed on an inclined plane, and the loaded part above, it will roll up-hill without assistance.

THE BALANCED STICK.

Procure a piece of wood, about nine inches in length, and about half an inch in thickness, and thrust into its upper end the blades of two pen-knives, on

either side one. Place the other end upon the tip of the fore-finger, and it will keep its place without falling.

THE CHINESE MANDARIN.

Construct out of the pith of the elder a little mandarin; then provide a base for it to sit in, like a kettle drum. Into this put some heavy substance, such as half a leaden bullet; fasten the figure to this, and in whatever position it may be placed, it will, when left to itself, immediately return to its upright position.

TO MAKE A SHILLING TURN ON ITS EDGE ON THE POINT OF A NEEDLE.

Take a bottle, with a cork in its neck, and place in it, in a perpendicular position, a middle-sized needle. Fix a shilling into another cork, by cutting a nick in it; and stick into the same cork two small table-forks, opposite each other, with the handles inclining outwards and downwards. If the rim of the shilling be now poised on the point of the needle, it may easily be made to spin round without falling, as the center of gravity is below the center of suspension.

THE DANCING PEA.

If you stick through a pea, or small ball of pith, two pins at right angles and defend the points with pieces of sealing-wax, it may be kept in equilibrio at a short distance from the end of a straight tube, by means of a current of breath from the mouth, which imparts a rotary motion to the pea.

OBLIQUITY OF MOTION.

Cut a piece of pasteboard into a circular shape, and describe on it a spiral line; cut this out with a pen-knife, and then suspend it on a large skewer or pin. If the whole be now placed on a warm stove, or over the flame of a candle or lamp, it will revolve with considerable velocity. The card, after being cut into the spiral, may be made to represent a snake or dragon, and when in motion will produce a very pleasing effect.

PNEUMATICS.

The branch of the physical sciences which relates to the air and its various phenomena is called Pneumatics. By it we learn many curious particulars. By it we find that the air has weight and pressure, color, density, elasticity, compressibility, and some other properties with which we shall endeavor to make the young reader acquainted by many pleasing experiments, earnestly impressing upon him to lose no opportunity of making physical science his study.

The common leather sucker by which boys raise stones will show the pressure of the atmosphere. It consists of a piece of soft but firm leather having a piece of string drawn through its center. The leather is made quite wet and pliable, and then its under part is placed on the stone and stamped down by the foot. This pressing excludes the air from between the leather and the stone, and by pulling the string a vacuum is left underneath its center; consequently the leather is firmly attached to the stone, which enables you to lift it.

WEIGHT OF THE AIR PROVED BY A PAIR OF BELLOWS.

Shut the nozzle and valve-hole of a pair of bellows, and after having squeezed the air out of them, if they are perfectly air-tight, we shall find that a very great force, even some hundreds of pounds, is necessary for separating the boards. They are kept together by the weight of the air which surrounds them in the same manner as if they were surrounded by water.

THE PRESSURE OF THE AIR SHOWN BY A WINE-GLASS.

Place a card on a wine-glass filled with water, then invert the glass; the water will not escape, the pressure of the atmosphere on the outside of the card being sufficient to support the water.

ANOTHER.

Invert a tall glass jar in a dish of water, and place a lighted taper under it; as the taper consumes the air in the jar, the water, from the pressure without,

rises up to supply the place of the oxygen removed by the combustion. In the operation of cupping the operator holds the flame of a lamp under a bell-shaped glass. The air within this being rarefied and expanded, a considerable portion is given off. In this state the glass is placed upon the flesh, and as the air within it cools it contracts, and the glass adheres to the flesh by the difference of the pressure of the internal and external air.

ELASTICITY OF THE AIR.

This can be shown by a beautiful philosophical toy, which may easily be constructed. Procure a glass jar and put water into it. Then mold three or four little figures in wax, and make them hollow within, and having each a minute opening at the heel, by which water may pass in and out. Place them in the jar, and adjust them by the quantity of water admitted to them, so that in specific gravity they differ a little from each other. The mouth of the jar should now be covered with a piece of skin or india-rubber, and then, if the hand be pressed upon the top or mouth of the jar, the figures will be seen to rise or descend as the pressure is gentle or heavy; rising and falling or standing still, according to the pressure made.

REASON FOR THIS.

The reason of this is, that the pressure on the top of the jar condenses the air between the cover and the water surface; this condensation then presses on the water below, and influences it through its whole extent, compressing also the air in the figures, forcing as much more water into them as to render them heavier than water, and therefore heavy enough to sink.

THE AIR-PUMP.

The time was, and that not very long ago, when the air-pump was only obtainable by the philosophical professor or by persons of enlarged means. But now, owing to our “cheap way of doing things,” a small air-pump may be obtained for about \$5, and we would strongly advise our young friends to procure one, as it will be a source of endless amusement to them; and, supposing that they take our advice, we suggest the following experiments.

The air-pump consists of a bell glass, called the receiver, and a stand upon which is a perforated plate. The hole in this plate is connected with two

pistons, the rods of which are moved by a wheel handle backwards and forwards, and thus pumps the air out of the receiver. When the air is thus taken out, a stop-cock is turned, and then the experiments may be performed.

Under the receiver of an air-pump, when the air has been thoroughly exhausted, light and heavy bodies fall with the same swiftness. Animals quickly die for want of air, combustion ceases, a bell sounds faint, and water and other fluids change to vapor.

TO PROVE THAT AIR HAS WEIGHT.

Take a florence flask, fitted up with a screw and fine oiled silk valve. Screw the flask on the plate of the air-pump, exhaust the air, take it off the plate, and weigh it. Then let in the air, and again weigh the whole, and it will be found to have increased by several grains.

TO PROVE AIR ELASTIC.

Place a bladder out of which all the air has apparently been squeezed under the receiver, upon it lay a weight, exhaust the air, and it will be seen that the small quantity of air left within the bladder will so expand itself as to lift the weight. Put a corked bottle into the receiver, exhaust the air, and the cork will fly out.

SOVEREIGN AND FEATHER.

Place a nicely-adjusted pair of forceps at the top of the receiver, communicating with the top of the outside through a hole, so that they may be opened by the fingers. Then place on each of the little plates a *sovereign* and a *feather*. Exhaust the air from the receiver: and having done so, detach the objects, so that they may fall. In the open air the sovereign will fall long before the feather, but in vacuo, as in the receiver now exhausted of its air, they will fall both together, and reach the bottom of the glass at the same instant.

AIR IN THE EGG.

Take a fresh egg, and cut off a little of the shell and film from its smaller end; then put the egg under a receiver, and pump out the air; upon which all the contents of the egg will be forced out by the expansion of the small bubble of air contained in the great end between the shell and the film.

THE DESCENDING SMOKE.

Set a lighted candle on the plate, and cover it with a tall receiver. The candle will continue to burn while the air remains, but when exhausted, will go out, and the smoke from the wick, instead of rising, will descend in dense clouds towards the bottom of the glass, because the air which would have supported it has been withdrawn.

THE SOUNDLESS BELL.

Set a bell on the pump-plate, having a contrivance so as to ring it at pleasure, and cover it with a receiver; then make the clapper sound against the bell, and it will be heard to sound very well; now exhaust the receiver of air, and then when the clapper strikes against the sides of the bell the sound can be scarcely heard.

THE FLOATING FISH.

If a glass vessel containing water, in which a couple of fish are put, be placed under the receiver, upon exhausting the air the fish will be unable to keep at the bottom of the glass owing to the expansion of the air within their bodies, contained in the air bladder. They will consequently rise and float, belly upwards, upon the surface of the water.

THE DIVING BELL.

The diving bell is a pneumatic engine, by means of which persons can descend to great depths in the sea, and recover from it valuable portions of wrecks and other things. Its principle may be well illustrated by the following experiment. Take a glass tumbler, and plunge it into the water with the mouth downwards, and it will be found that the water will not rise much more than half way in the tumbler. This may be made very evident if a piece of cork be suffered to float inside the glass on the surface of the water. The air within the tumbler does not entirely exclude the water,

because air is elastic, and consequently compressible, and hence the air in the tumbler is what is called condensed. The diving bell is formed upon the above principle; but instead of being glass it is a wooden or metal vessel, of very large dimensions, so as to hold three or four persons, who are supplied with air from above by means of powerful pumps, whilst the excess of air escapes at the bottom of the bell.

EXPERIMENTS.

1. Place a cylinder of strong glass, open at both ends, on the plate of the air-pump, and put your hand on the other end, and you will of course be able to remove it at pleasure. Now exhaust the air from the interior of the cylinder, and at each stroke of the pump you will feel your hand pressed tighter and tighter on the cylinder, until you will not be able to remove it: as soon as the air is again admitted to the interior of the cylinder, the pressure within will be restored, and the hand again be at liberty.

2. Tie a piece of moistened bladder very firmly over one end of a similar glass cylinder, and place the open end on the plate of the pump. As soon as you begin to exhaust the air from the interior, the bladder, which was previously quite horizontal, will begin to bulge inwards, the concavity increasing as the exhaustion proceeds, until the bladder, no longer able to bear the weight of the superincumbent air, breaks with a loud report.

3. The elasticity of air, or indeed of any gaseous body, may be shown by introducing under the air-pump receiver a bladder containing a very small quantity of air, its mouth being closely tied. As you exhaust the air from the receiver, that portion contained in the bladder being no longer pressed upon by the atmosphere, will gradually expand, distending the bladder until it appears nearly full: on readmitting the air into the receiver, the bladder will at once shrink to its former dimensions.

A shriveled apple placed under the same conditions will appear plump when the air is removed from the receiver, and resume its former appearance on the readmission of the air.

4. There is a very pretty apparatus made for the purpose of showing the pressure of the atmosphere, consisting of a hollow globe of brass, about three inches in diameter, divided into two equal parts, which fit very accurately together. It is furnished with two handles; one of them screwed

into a hollow stem, communicating with the interior of the globe, and fitting on to the air-pump; the other is attached to a short stem on the opposite side of the globe. In the natural state the globe may easily be separated into its two hemispheres by one person pulling the handles, but after the air has been exhausted from the interior it requires two very strong men to separate the parts, and they will often fail to do so. By turning the stop-cock, and readmitting the air into the interior of the globe, it will come asunder as easily as at first.

We are indebted to the weight of the atmosphere for the power we possess of raising water by the common pump; for the piston of the pump withdrawing the air from the interior of the pipe, which terminates in water, the pressure of the atmosphere forces the water up the pipe to supply the place of the air withdrawn. It was soon found, however, that when the column of water in the pipe was more than thirty feet high, the pump became useless, for the water refused to rise higher. Why? It was found that a column of water about thirty feet high exerted a pressure equal to the weight of the atmosphere, thus establishing an equilibrium between the water in the pipe and the atmospheric pressure.

This is the principle on which the barometer, or *measurer of weight*, as its name imports, is constructed. The metal Mercury is about thirteen and a half times heavier than water; consequently, if a column of water thirty *feet* high balances the pressure of the atmosphere, a column of mercury thirty *inches* high ought to do also—and this is in fact the case. If you take a glass tube nearly three feet long, and closed at one end, and fill it with mercury; then, placing your finger on the open end, invert the tube into a basin or saucer containing some of the same metal; upon removing your finger (which must be done carefully, while the mouth of the tube is completely covered by the mercury), it will be seen that the fluid will fall a few inches, leaving the upper part of the tube empty. Such a tube with a graduated scale attached is in truth a barometer, and as the weight of the atmosphere increases or decreases, so the mercury rises or falls in the tube. This instrument is of the greatest value to the seaman, for a sudden fall of the barometer will often give notice of an impending storm when all is fine and calm, and thus enable the mariner to make the preparations necessary to meet the danger.

It was discovered by an Italian philosopher named Torricelli, and from him the vacuum formed in the upper end of the tube above the surface of the mercury has been called the Torricellian vacuum. It is by far the most perfect vacuum that can be obtained, containing necessarily nothing but a minute quantity of the vapor of mercury.

EXPERIMENT.

Pass a little ether through the mercury in the tube, and as soon as it reaches the empty space it will boil violently, depressing the mercury, until the pressure of its own vapor is sufficient to prevent its ebullition. If you now cool the upper part of the tube, so as to condense the vapor, the pressure being thus removed, the ether will again begin to boil, and so alternately, as often as you please. In order to show this fact with effect, the bore of the tube should not be less than half an inch in diameter.

EXPERIMENT.

To show that the heat abstracted by the boiling of one liquid will freeze another, fill a tall narrow glass about half full of cold water (the colder the better), and place in it a thin glass tube containing some ether. Put them under the receiver of an air-pump. As you exhaust the air, the ether will begin to boil, until at length, by continuing the exhaustion, the water immediately surrounding the tube of ether will freeze, and a tolerably large piece of ice may thus be obtained.

Ether evaporates so rapidly even under the pressure of the atmosphere, that a small animal, such as a mouse, may be actually frozen to death by constantly dropping ether upon it. If poured on the hand, it produces a degree of cold that soon becomes, to say the least, unpleasant.

EXPERIMENT.

Place a flat saucer containing about a pound of oil of vitriol under the receiver of the air-pump, and set in it a watch glass containing a little water, supported on a stand with *glass* legs. Exhaust the receiver, when the water will evaporate, but without boiling; and the vapor being absorbed as it forms by the oil of vitriol, the vacuum is preserved, and the evaporation

continues, until the vapor has abstracted so much caloric from the remainder of the water that it is all at once converted into ice.

In most elementary works on chemistry may be found a long table of freezing mixtures, as they are called, some with and others without ice or snow. We have selected a few from each division.

WITH ICE OR SNOW.

{	Snow or powdered ice	2 parts.
{	Powdered common salt	1 “
	Snow	5 “
{	Powdered common salt	2 “
	Powdered sal ammoniac	1 “
{	Snow	3 “
{	Dilute sulphuric acid	2 “
{	Snow	2 “
{	Crystallized muriate of lime	3 “

WITHOUT SNOW OR ICE.

{	Sulphate of soda	3 parts.
{	Dilute nitric acid	2 “
{	Nitrate of ammonia	1 “
{	Water	1 “
{	Phosphate of soda	2 “
{	Dilute nitric acid	1 “
{	Sulphate of soda	2 “
{	Muriatic acid	1 “

The effects of most of these mixtures may be considerably increased by previously cooling the ingredients *separately* in other freezing mixtures.

In connection with this branch of science, and especially with chemistry, the youthful philosopher should practice the art of decanting air from one jar to another standing over water, beginning by passing it from a small to a larger jar, then with two of equal size; and when he can accomplish the transfer

without permitting even one bubble to escape, he may essay the much more difficult task of transferring the air from a large to a smaller jar.

He should also practice using the blowpipe until he can keep up a steady and uninterrupted flame for ten minutes or a quarter of an hour, without stopping for breath. It is quite possible to replenish *wind* in the mouth, which alone ought to be used, without interrupting the breathing for an instant, but it requires some practice.

HOW TO BECOME AN OPTICIAN.

Optics is the science of *light* and *vision*. Concerning the nature of light, two theories are at present very ably maintained by their respective advocates. One is termed the Newtonian theory, and the other the Huygenean. The Newtonian theory considers light to consist of inconceivably small bodies emanating from the sun, or any other luminous body. The Huygenean conceives it to consist in the undulations of a highly elastic and subtle fluid, propagated round luminous centers in spherical waves, like those arising in a placid lake when a stone is dropped into the water.

LIGHT AS AN EFFECT.

Light follows the same laws as gravity, and its intensity or degree decreases as the square of the distance from the luminous body increases. Thus, at the distance of two yards from a candle we shall have four times less light than we should have were we only one yard from it, and so on in the same proportion.

REFRACTION.

Bodies which suffer the rays of light to pass through them, such as air, water, or glass, are called refracting media. When rays of light enter these, they do not proceed in straight lines, but are said to be refracted, or bent out of their course. But if the ray falls perpendicularly on the glass, there is no refraction, and it proceeds in a direct line; hence, refraction only takes place when rays fall obliquely or aslant on the media.

THE INVISIBLE COIN MADE VISIBLE.

If a coin be placed in a basin, so that on standing at a certain distance it be just hid from the eye of an observer by the rim or edge of the basin, and then water be poured in by a second person, the first keeping his position; as the water rises the coin will become visible, and will appear to have moved from the side to the middle of the basin.

THE MULTIPLYING GLASS.

The multiplying glass is a semicircular piece of glass cut into facets or distinct surfaces; and in looking through it we have an illustration of the laws of refraction, for if a small object, such as a fly, be placed at the further end, a person will see as many flies as there are surfaces or facets on the glass.

TRANSPARENT BODIES.

Transparent bodies, such as glass, may be made of such form as to cause all the rays which pass through them from any given point to meet in any other given point beyond them, or which will disperse them from the given point. These are called lenses, and have different names according to their form. 1. Is called the plano-convex lens. 2. Plano-concave. 3. Double convex. 4. Double concave. 5. A meniscus, so called from its resembling the crescent moon.

THE PRISM.

The prism is a triangular solid of glass, and by it the young optician may decompose a ray of light into its primitive and supplementary colors, for a ray of light is of a compound nature. By the prism the ray is divided into its three primitive colors, blue, red, and yellow; and their four supplementary ones, violet, indigo, green, and orange. The best way to perform this experiment is to cut a small slit in a window-shutter, on which the sun shines at some period of the day, and directly opposite the hole place a prism; a beam of light in passing through it will then be decomposed, and if let fall upon a sheet of white paper, or against a white wall, the seven colors of the rainbow will be observed.

COMPOSITION OF LIGHT.

The beam of light passing through the prism is decomposed, and the spaces occupied by the colors are in the following proportions: Red, 6; orange, 4; yellow, 7; green, 8; blue, 8; indigo, 6; violet, 11. Now, if you paste a sheet of white paper on a circular piece of board about six inches in diameter, and divide it with a pencil into fifty parts, and paint colors in them in the proportions given above, painting them dark in the center parts, and gradually fainter at the edges, till they blend with the one adjoining. If the

board be then fixed to an axle, and made to revolve quickly, the colors will no longer appear separate and distinct, but becoming gradually less visible they will ultimately appear *white*, giving this appearance to the whole surface of the paper.

A NATURAL CAMERA OBSCURA.

The human eye is a camera obscura, for on the back of it on the retina every object in a landscape is beautifully depicted in miniature. This may be proved by the

BULLOCK'S EYE EXPERIMENT.

Procure a fresh bullock's eye from the butcher, and carefully thin the outer coat of it behind: take care not to cut it, for if this should be done the vitreous humor will escape, and the experiment cannot be performed. Having so prepared the eye, if the pupil of it be directed to any bright objects, they will appear distinctly delineated on the back part precisely as objects appear in the instrument we are about to describe. The effect will be heightened if the eye is viewed in a dark room with a small hole in the shutter, but in every case the appearance will be very striking.

THE CAMERA OBSCURA.

This is a very pleasing and instructive optical apparatus, and it may be easily made by the young optician. Procure an oblong box, about two feet long, twelve inches wide, and eight high. In one end of this a tube must be fitted containing a lens, and be made to slide backwards and forwards so as to suit the focus. Within the box should be a plain mirror reclining backwards from the tube at an angle of forty-five degrees. At the top of the box is a square of unpolished glass, upon which from beneath the picture will be thrown, and may be seen by raising the lid. To use the camera place the tube with the lens on it opposite to the object, and having adjusted the focus, the image will be thrown upon the ground-glass as above stated, where it may be easily copied by a pencil or in colors.

The camera obscura used in a public exhibition is a large wooden box stained black in the inside, and capable of containing from one to eight persons. It contains a sliding piece, having a sloping mirror and a double

convex lens which may with the mirror be slid up or down so as to accommodate the lens to near and distant objects. When the rays proceeding from an object without fall upon the mirror they are reflected upon the lens, and brought to fall on the bottom of the box, or upon a table placed horizontally to receive them, which may be seen by the spectator.

THE MAGIC LANTERN.

This is one of the most pleasing of all optical instruments, and it is used to produce enlarged pictures of objects, which being painted on a glass in various colors are thrown upon a screen or white sheet placed against the wall of a large room. It consists of a sort of tin box, within which is a lamp, the light of which passes through a great plano-convex lens fixed in the front. This strongly illuminates the objects which are painted on the slides or slips of glass, and placed before the lens in an inverted position, and the rays passing through them and the lens fall on a sheet or other white surface, placed to receive the image. The glasses on which the figures are drawn are inverted, in order that the images of them may be erect.

PAINTING THE SLIDES.

The slides containing the objects usually shown in a magic lantern, are to be bought at opticians with the lantern, and can be procured cheaper and better in this way than by any attempt at manufacturing them. Should, however, the young optician wish to make a few slides of objects of particular interest to himself, he may proceed as follows:

Draw first on paper the figures you wish to paint, lay it on the table, and cover it over with a piece of glass of the above shape; now draw the outlines with a fine camel's hair pencil in black paint mixed with varnish, and when this is dry fill up the other parts with the proper colors, shading with bister also mixed with varnish. The transparent colors are alone to be used in this kind of painting.

TO EXHIBIT THE MAGIC LANTERN.

The room for the exhibition ought to be large, and of an oblong shape. At one end of it suspend a large sheet so as to cover the whole of the wall. The company being all seated, darken the room, and placing the lantern with its

tube in the direction of the sheet, introduce one of the slides into the slit, taking care to invert the figures; then adjust the focus of the glasses in the tube by drawing it in or out as required, and a perfect representation of the object will appear.

EFFECTS OF THE MAGIC LANTERN.

Most extraordinary effects may be produced by means of the magic lantern; one of the most effective of which is a

TEMPEST AT SEA.

This is effected by having two slides painted, one with the tempest as approaching on one side, and continuing in intensity till it reaches the other. Another slide has ships painted on it, and while the lantern is in use, that containing the ships is dexterously drawn before the other, and represents *ships in the storm*.

The effects of sunrise, moonlight, starlight, etc., may be imitated also, by means of double slides, and figures may be introduced sometimes of *fearful* proportions.

Heads may be made to nod, faces to laugh; eyes may be made to roll, teeth to gnash; crocodiles may be made to swallow tigers; combats may be represented; but one of the most instructive uses of the slides is to make them illustrative of astronomy, and to show the rotation of the seasons, the cause of eclipses, the mountains in the moon, spots on the sun, and the various motions of the planetary bodies, and their satellites.

THE PHANTASMAGORIA.

Between the phantasmagoria and the magic lantern there is this difference: in common magic lanterns the figures are painted on transparent glass, consequently the image on the screen is a circle of light having figures upon it; but in the phantasmagoria all the glass is made opaque, except the figures, which, being painted in transparent colors, the light shines through them, and no light can come upon the screen except that which passes through the figure.

There is no sheet to receive the picture, but the representation is thrown on a thin screen of silk or muslin placed between *the spectators and the lantern*. The images are made to appear approaching and receding by removing it further from the screen, or bringing it nearer to it. This is a great advantage over the ordinary arrangements of the magic lantern, and by it the most astonishing effects are often produced.

DISSOLVING VIEWS.

The dissolving views, by which one landscape or scene appears to pass into the other while the scene is changing, are produced by using two magic lanterns placed side by side, and that can be a little inclined towards each other when necessary, so as to mix together the rays of light proceeding from the lenses of each, which produces that confusion of images, in which one view melts as it were into the other, which gradually becomes clear and distinct; the principle being the gradual extinction of one picture, and the production of another.

HOW TO RAISE A GHOST.

The magic lantern, or phantasmagoria, may be used in a number of marvelous ways, but in none more striking than in raising an apparent specter. Let an open box, about three feet long, a foot and a half broad, and two feet high, be prepared. At one end of this place a small swing dressing-glass, and at the other let a magic lantern be fixed with its lenses in a direction towards the glass. A glass should now be made to slide up and down in the groove, to which a cord and pulley should be attached, the end of the cord coming to the back part of the box. On this glass the most hideous specter that can be imagined may be painted, but in a squat or contracted position, and when all is done, the lid of the box must be prepared by raising a kind of gable at the end of the box, and in its lower part an oval hole should be cut sufficiently large to suffer the rays of light reflected from the glass to pass through them. On the top of the box place a chafing-dish, upon which put some burning charcoal. Now light the lamp in the lantern, sprinkle some powdered camphor or white incense on the charcoal, adjust the slide on which the specter is painted, and the image will be thrown upon the smoke. In performing this feat the room must be

darkened, and the box should be placed on a high table, that the hole through which the light comes may not be noticed.

THE THAUMATROPE.

This word is derived from two Greek words, one of which signifies *wonder*, and the other *to turn*. It is a very pretty philosophical toy, and is founded upon the principle in optics that an impression made upon the retina of the eye lasts for a short interval after the object which produced it has been withdrawn. The impression which the mind receives lasts for about the eighth part of a second, as may be easily shown by whirling round a lighted stick, which if made to complete the circle within that period, will exhibit not a fiery point, but a fiery circle in the air.

THE BIRD IN THE CAGE.

Cut a piece of cardboard of the size of a penny piece, and paint on one side a bird, and on the other a cage; fasten two pieces of thread, one on each side at opposite points of the card, so that the card can be made to revolve by twirling the threads with the finger and thumb: while the toy is in its revolution, the bird will be seen within the cage. A bat may in the same manner be painted on one side of the card, and a cricketer upon the other, which will exhibit the same phenomenon, arising from the same principle.

CONSTRUCTION OF THE PHANTASMASCOPE.

The above named figure is a Thaumatrope, as much as the one we are about to describe, although the term Phantasmascope is generally applied to the latter instrument; which consists of a disc of darkened tin-plate, with a slit or narrow opening in it, about two inches in length. It is fixed upon a stand, and the slit placed upwards, so that it may easily be looked through. Another disc of pasteboard, about a foot in diameter, is now prepared and fixed on a similar stand, but with this difference, that it is made to revolve round an axis in the center. On this pasteboard disc, paint in colors a number of frogs in relative and progressive positions of leaping; make between each figure a slit of about a quarter of an inch deep: and when this second disc is made to revolve at a foot distance behind the first, and the eye is placed near the slit, the whole of the figures, instead of appearing to

revolve with the disc, will all appear in the attitudes of leaping up and down, increasing in agility as the velocity of the motion is increased. It is necessary, when trying the effect of this instrument, to stand before a looking-glass, and to present the painted face of the machine toward the glass.

A very great number of figures may be prepared to produce similar effects—horses with riders in various attitudes of leaping, toads crawling, snakes twisting and writhing, faces laughing and crying, men dancing, jugglers throwing up balls, etc.; all of which, by the peculiar arrangement above detailed, will seem to be in motion. A little ingenuity displayed in the construction and painting of the figures upon the pasteboard disc will afford a great fund of amusement.

CURIOUS OPTICAL ILLUSIONS.

One of the most curious facts relating to the science of vision is the absolute insensibility of a certain portion of the retina to the impression of light, so that the image of any object falling on that point would be invisible. When we look with the right eye, this point will be about fifteen degrees to the right of the object observed, or to the right of the axis of the eye, or the point of most distinct vision. When looking with the left eye, the point will be as far to the left. The point in question is the basis of the optic nerve, and its insensibility to light was first observed by the French philosopher, Mariotte. This remarkable phenomenon may be experimentally proved in the following manner:—

Place on a sheet of writing-paper, at the distance of about three inches apart, two colored wafers; then, on looking at the left-hand wafer with the right eye, at the distance of about a foot, keeping the eye straight above the wafer, and both eyes parallel with the line which forms the wafers, the left eye being closed, the right-hand wafer will become invisible; and a similar effect will take place if we close the right eye, and look with the left.

ANOTHER.

Cut a circular piece of white paper, about two inches in diameter, and affix it to a dark wall. At the distance of two feet on each side, but a little lower, make two marks; then place yourself directly opposite the paper, and hold

the end of your finger before your face, so that when the right eye is open it shall conceal the mark on your left, and when the left eye is open the mark on your right. If you then look with both eyes at the end of your finger the paper disc will be invisible.

ANOTHER.

Fix a similar disc of paper, two inches in diameter, at the height of your eye on a dark wall; a little lower than this, at the distance of two feet on the right hand, fix another of about three inches in diameter; now place yourself opposite the first sheet of paper, and, shutting the left eye, keep the right eye still fixed on the first object, and when at the distance of about ten feet, the second piece of paper will be invisible.

THE PICTURE IN THE AIR.

One of the numerous optical illusions which have from time to time been evolved by scientific minds, is that of making an image or picture appear in the air. This is produced by means of a mirror, and an object in relief, upon which a strong light is thrown—the mirror being set at such an angle as to throw up the reflection of the image to a certain point in the view of the spectator. This illusion is produced as follows: Let a screen be constructed in which is an arched aperture, the center of which may be five feet from the floor; behind the screen is placed a large mirror of an elliptical form. An object is now placed behind the screen, upon which the light of a strong lamp is thrown from a point above the mirror, and is received by the mirror and reflected to the center of the arched cavity in the screen, where it will appear to the spectator. Care should be taken to place the image in an inverted position, and the light, which must be very powerful, should be so placed that none of it may reach the opening.

TO SHOW THAT RAYS OF LIGHT DO NOT OBSTRUCT EACH OTHER.

Make a small hole in a sheet of pasteboard, and placing it upright before three candles, placed closely together, it will be found that the images of all the candle flames will be formed separately on a piece of paper, laid on the

table to receive them. This proves that the rays of light do not obstruct each other in their progress, although all cross in passing through the hole.

OPTICS OF A SOAP-BUBBLE.

If a soap-bubble be blown up, and set under a glass, so that the motion of air may not affect it, as the water glides down the sides and the top grows thinner, several colors will successively appear at the top, and spread themselves from thence down the sides of the bubble, till they vanish in the same order in which they appeared. At length a black spot appears at the top, and spreads till the bubble bursts.

THE KALEIDOSCOPE.

If any object be placed between two plane mirrors, inclined towards each other at an angle of thirty degrees, three several images will be perceived in the circumference of a circle. On this principle is formed the kaleidoscope, invented by Sir David Brewster, and by means of which the reflected images viewed from a particular point exhibit symmetrical figures, under an infinite arrangement of beautiful forms and colors. The kaleidoscope may be bought at any novelty store, but it is requisite that every young person should be able to construct one for himself. He must, therefore, procure a tube of tin or paper, of about ten inches in length, and two and a half or three inches in diameter. One end of this should be stopped up with tin or paper, securely fastened, in which is to be made a hole, about the size of a small pea, for the eye to look through. Two pieces of well-silvered looking-glass are now to be procured; they must be not quite so long as the tube, and they should be placed in it lengthways, at an angle of 60 degrees, meeting together in a point, and separating to an angle wide enough to insert the third piece; the polished surfaces looking inwards. A circular piece of the glass is now to be laid on the top of the edges of the reflectors; which, by their not being quite so long as the tube, will allow room for its falling in, and it will be supported by the edges of the tube, which may be slightly bent over, to prevent the glass from falling out. This having been done, now proceed to make the “cap” of the instrument. A rim of tin or pasteboard must be cut, so as to fit over the glass end of the tube; and in this, on the outer side, a piece of ground glass must be fastened, so that the whole may fit on the tube like the lid of a pill-box. Then, before putting it on, obtain

some small pieces of broken glass of various colors, beads, little strips of wire, or any other object, and place them in the cap; and by passing it over the end, so that the broken glass, etc., has free motion, the instrument is complete. To use it, apply the eye to the small hole, and, on turning it, the most beautiful forms will appear, in the most wonderful combinations.

The following curious calculation has been made of the number of changes this instrument will admit of. Supposing it to contain 20 small pieces of glass, and that you make 10 changes in a minute, it will take an inconceivable space of time, *i.e.* 462,880,899,576 years, and 360 days, to go through the immense number of changes of which it is capable.

SIMPLE SOLAR MICROSCOPE.

Having made a circular hole in a window-shutter, about three inches in diameter, place in it a glass lens of about twelve inches focal distance. To the inside of the hole adapt a tube, having at a small distance from the lens a slit, capable of receiving one or two very thin plates of glass, to where the object to be viewed must be affixed by means of a little gum-water exceedingly transparent. Into this tube fit another, furnished at its extremity with a lens half-an-inch focal distance. Place a mirror before the hole of the window-shutter on the outside, in such a manner as to throw the light of the sun into the tube, and you will have a solar magic lantern.

The method of employing this arrangement of lenses for microscopic purposes is as follows:—Having darkened the room, and by means of the mirror reflected the sun's rays on the glasses in a direction parallel to the axis, place some small object between the two movable plates of glass, or affix it to one of them with very transparent gum-water, and bring it exactly into the axis of the tube; if the movable tube be then pushed out or drawn in, till the object be a little beyond the focus, it will be seen painted very distinctly on a card, or piece of white paper, held at a proper distance, and will appear to be greatly magnified. A small insect will appear as a large animal, a hair as big as a walking-stick, and the almost invisible eels in paste or vinegar as large as common eels.

THE MICROSCOPE.

At any time of the year or hour of the day there are few pursuits more interesting, and at the same time instructive, than the study of Nature by means of the microscope.

All of us must admire the more than awful grandeur of that universe whereof we form so infinitesimal a part, wherein the stars are scattered as the sand on the sea-shore, and every star a sun, the center of a system of orbs too distant for the eye of man to perceive. Looking at our nearest planet, and observing on her face vast mountain-chains, ravines into which the light of the sun can never penetrate, and volcanoes whose craters are so wide that they would take in the whole of New York, the whole of Philadelphia, and all the country between them, we can judge by analogy of the unseen wonders which must exist in the world beyond our ken.

But to him who can read Nature rightly, the microscope is a teacher as grand as its sister instrument, and the awful magnificence of Nature is as evident in a midge's wing as in the more patent glories of the sun, moon, and stars. In the following pages we hope to put the readers of this book in the way to read their microscope rightly—possibly to make it—and to show that much can be done with small means when “there's a will,” and to indicate to them that objects of no small interest can be found without stirring from the room in which we sit, or even from the table on which our microscope is placed.

Some of our readers may say, when they read the heading of this paper, that they should like a microscope very much, but that they have no money to buy it, and that their parents cannot afford one.

This is just the feeling which we used to have when a boy, for in those day microscopes were microscopes indeed, and you had your choice between a little instrument, with a series of brass cups, having glasses in them, which magnified slightly but defined clearly, or a great composition of brass and iron, looking like a rocket-tube, with an eye-piece at one end and a glass shot at the other. It was very costly, very imposing, and magnified very highly; but it strained the eyes painfully, had no defining capacities, and

made all the objects look as if they were seen through a thick fog. Practically, therefore, the former was the only instrument that was available.

A still more useful instrument, however, was that which can always be obtained for a dollar or so, and which is now made wonderfully cheap and wonderfully good; we mean the double or treble pocket-lens. So we say, if you cannot afford a really good microscope, do not waste your money upon inferior and pretentious instruments, but get a sound pocket-lens.

It has a thousand advantages. It is portable, and is even more useful in the fields than in the house. It defines very clearly, and needs little trouble in manipulation. We need not say how difficult is the task of getting a complicated instrument to define properly, how impossible with a bad one. The object and the glass can be held in any light,—a matter of no small consideration when examining anything new, and trying to make out its structure. It is not easily put out of order, and if treated with the most ordinary care, will last for a lifetime.

You can push it under water, and it will magnify as well as in the air; and if you are wandering on the river-side, you can lie down on the bank, dip the upper part of your head in water, together with the glass, and watch carefully the sub-aquatic objects without removing them. The water will not hurt the eye in the least, though a non-swimmer may perhaps find a little difficulty in his first attempt. It makes a good burning-glass, should fire be needed, and no other means of procuring a spark be at hand. It can be used so as to show the principle of a camera obscura, and to illustrate the manner in which photographic portraits are taken. It can be made into an admirable dissecting microscope, and needs scarcely any practice in the manipulation. These are some of its advantages, and there are many others which need not be mentioned.

Even if you should be able to procure a good microscope, get a pocket-lens as well, for you will want them both, and we may say that the most practiced microscopists, and those who are possessors of the most elaborate instruments, are the very men who are the most certain to have a pocket-lens about them, and to use it most frequently. Practice well with the pocket-lens before you meddle with the compound microscope. You will waste no time, but will rapidly gain by it; for you will be learning the

rudiments of a new science, and laying a solid foundation on which to build.

One or two practical remarks on the proper handling of the pocket-lens may be of use. Do not always employ the same eye in looking through the lens, but use the eyes alternately. There is always a temptation to employ the same eye, which receives a kind of training in vision; but it is a temptation always to be resisted. With some persons the right eye is most in favor, and with others the left; and when the favorite eye gets all the work, it too frequently suffers. Whether you look with the right or the left eye, *keep both eyes open*.

At first the beginner will find a little difficulty in restricting his vision to one eye while the other remains open, just as a beginner on the piano-forte feels himself puzzled when he tries to make his right hand go one way and his left hand another; but in either case a little practice and plenty of perseverance are sure to overcome all obstacles, and in a wonderfully short time the difficulty will not only be overcome, but forgotten.

We speak here with some feeling, because, while engaged on a work on the microscope, we were necessarily obliged to work much at night, and inadvertently employed the left eye more than the right; the consequence of which imprudence was that we have been obliged ever since that time to give the left eye perfect rest, as far as artificial vision goes, and, except when looking through a binocular instrument, we have not ventured to use it either to a microscope or a telescope. The vision accommodates itself to circumstances with wonderful ease, and the observer learns the curious art of cutting off all communication between the unused eye and the brain; so that, although the objects around may imprint themselves upon the retina, the mind is as totally unconscious of them as if they had no existence.

If possible, always examine an object *without removing it*, as thereby you see it as it is, without altering any of the conditions with which it is surrounded. Should this not be practicable, take the object to be viewed in the left hand and the lens in the right. Place the wrists of the two hands together, and then you will find that one supports the other, and that the lens can be held in the proper focus without the least difficulty. After you have used the lens for some little time, you will learn to hit upon the right focus almost to a hair's breadth,—so as to lose no time, a matter of some

importance when a living creature is to be examined, especially if it be in motion.

We are now about to suggest a very simple piece of mechanism, by which the pocket-lens can be converted into a microscope that will serve for dissection and many other purposes.

Melt three or four pounds of lead in an iron ladle, and make a mold, consisting of a hollow hemisphere of paper or cardboard, through the center of which an iron rod has been passed. The hollow of the paper should resemble an ordinary saucer. Pour the lead into the saucer, and let it cool. The paper mold will be scorched by the heat and rendered useless, but an outer coating of lead will be cool and hard before the paper is quite destroyed. Next take a piece of stout brass wire and a wine-cork; twist the wire round the cork several times; cut off one end close to the cork; sharpen the other, and turn it up.

Bore a hole through the cork, just large enough to allow the upright rod to slip through it, and there is the “stand” of your microscope. Now take your pocket-lens, and get an optician to bore a hole through one end of it, just large enough to receive the upturned end of the wire; slip the lens on the wire, and the microscope is complete.

The cork, though grasping the upright stem with tolerable firmness, can be slid up and down so as to insure the correct focus, and can be pushed aside whenever the object has to be viewed with the naked eye, and must not be removed from its place. This instrument is a capital one for dissecting purposes, and will answer quite as well as those expensive affairs that are to be purchased in the shops.

If the object be transparent, and requires to be seen by transmitted light, the following plan will answer:—Take a thin piece of wood, cut or punch a round hole out of the middle, and support it on four legs. Wires or wooden pegs fixed in corks will answer the purpose well, and if the corks be glued to the corners of the board, the legs can be inserted or removed at pleasure. The wood of which cigar-boxes are made will answer the purpose very well. Its dimensions should be about three inches in length by two in width. Now buy one of the doll’s looking-glasses that are sold for a penny, and put it under the stand. Lay a flat piece of glass over the hole, place the object upon it, and direct the light through it by means of the mirror below. If such

a mirror cannot be obtained, it is easy enough to make one, by mounting a piece of looking-glass in a cork frame, and making it swing on pivots, like the glasses of our dressing-rooms.

The young microscopist must remember that when he is examining any object by transmitted light, he must arrange it as flatly as possible on the glass. In many cases, a still neater manipulation is required—as, for example, when the petals of flowers are under examination. Thin glass is to be purchased at any optician's, and if cut in squares, instead of circles, is very much cheaper, and quite as useful for all practical purposes. Lay the petal on the glass plate, place a piece of the thin glass upon it, and press it gently while examining it. If it still remains thick and dull, put a drop of pure water on the petal, and replace the thin glass, when the structure will almost invariably be detected.

Everything depends on the proper management of the object and the arrangement of the light. Some opaque objects can be seen best by direct light, and others by transmitted light. If a leaf be examined, particularly if it be a thick and heavy one, like that of the ivy, the upper and lower membranes must be stripped apart—a task which is easily performed by tearing a small slit, and then ripping it smartly across. A pair of forceps will be required for this and other delicate work, and may be obtained at a cheap rate. Care must be taken to keep the points exactly even, and if at any time one of them appears to be shorter than the other, they should be rubbed on a hone until they are brought perfectly level.

These should be made of steel; but the young microscopist will find that a second pair, made of brass, and much rougher in finish, are invaluable aids as he takes his walks into the country. By their aid he can pick up minute objects, draw insects out of crevices without damaging them, and pluck the tiniest flowers without harming their petals. They can be carried in the waistcoat pocket, and the cost is sixpence. Any lad who knows how to handle solder can make a pair for himself in a few minutes.

A penknife with one blade kept scrupulously sharp is essential, and we have found an old lancet of the greatest service. Lancets have gone so much out of fashion, that the second-hand instrument shops abound with them. We did not allow our own lancet to be shut up, but removed the blade from the

tortoise-shell handle, and fixed it upon a wooden handle, about four inches in length, so that it looked very clumsy, but was extremely useful.

Two pairs of scissors are needful,—one very fine and the other moderately strong. Both pairs, however, must have very short blades and very long handles, and the scissors such as ladies use are of very little use, the short handles causing the fingers of the right hand to shade the object. As to the fine pair, it is hardly possible to have the handles too long or the blades too short; for if the points can be separated a quarter of an inch, nothing more is needed. If a pair of bent scissors can also be obtained, they are extremely pleasant to work with, and save much trouble.

Pill-boxes of various sizes are of very great service to the microscopist. We always have them arranged in “nests,” *i. e.*, six or seven inside each other, so that space is greatly economized, as long as they are not in absolute use. All delicate objects should be placed in separate boxes, and the predaceous insects must be treated in the same manner, or they will certainly destroy one another, or, at all events, inflict such injuries as will make them useless for microscopic purposes.

When the insects are to be killed on the spot, we employ another and a very simple plan.

We take one of the old-fashioned wooden lucifer-match boxes, bore a hole in the lid, and push through the hole a swan-quill, or the barrel of one of the swan-quill steel pens. A glass tube is still better, but is too fragile. Beeswax is tightly worked into the junction of the tube with the wood, so as to make it as nearly air-tight as possible. A cork stopper is then cut to fit the tube. When this is finished, we take the smallest-sized pill-box, bore a number of holes in it with a red-hot needle, place a little piece of solid ammonia within it, and inclose it in the lucifer-box. Its effects are almost instantaneous; for scarcely has the insect touched the bottom of the box before it is helpless, and in a very few moments it is quite dead, so powerful is ammonia towards insects. The reader will of course understand that the pill-boxes must never have been used for pills, and that the match-box must be carefully cleaned before employing it in the microscopic service. Moreover, any boxes that have been used for insects become useless, inasmuch as the scales always fall from the wings, and cling to the sides of the box, so as to mix with succeeding objects, and very much puzzle the observer.

Aquatic and marine objects require bottles, and, as a general rule, these bottles ought always to have wide mouths. Indeed, if there be no shoulder at all, their purpose will be better served, as a small object is very apt to be caught under the shoulder, and to give much trouble before it can be removed without injury. Wide and short test-tubes answer admirably for collecting; and it will always be advisable to have a few small test-tubes ready fitted with corks, for the purpose of isolating those specimens which might receive or cause injury by being mixed with others.

To remove minute objects from one vessel into another is a very easy process. Take a glass tube, mark off a portion about eight inches in length, cut a little notch with a file, and bend it smartly, when it will break neatly across, without leaving points or having the regularity of its ends injured by gaps. Turn each end round and round in the flame of the spirit-lamp, and you have an ordinary "pipette." The object of placing the ends of the tube in the flame is to render the edges quite smooth and rounded.

Now mark off the same length of tube, and place the marked portion in the flame, taking care to warm it well first, lest the sudden heat should crack the glass. Keep it continually turning between the fingers, and when it is quite soft, and of a fine red heat, draw the hands smartly apart, and you will produce a couple of tubes tapering to very fine points. Break off the tapering portions at any convenient point, round the edges as before, and you will then have pipettes suitable for small objects. As there are many specimens, especially the smaller animalculæ, which have a habit of retiring into the remotest corner, it is necessary to bend another pipette, so as to follow them. For our own part, we prefer the pipette to be bent nearly to a right angle.

The mode of using these simple instruments is as follows:—Place the forefinger or thumb firmly on the large end, and push the point under water. When the opening is close to the sought-for object, lift the finger suddenly, and admit the air into the tube. The water will immediately rush in at the lower end, and if the orifice has been properly directed, will carry the object into the tube. The finger is again applied to the mouth of the tube, and the object can be then carried off.

As with the pocket-lens almost every object is to be viewed by means of direct light, the young observer will find himself much aided by a suitable

background. Any small object, such as a minute insect, a seed, or a hair, becomes very indistinct if held up against the light, or even when viewed against a broken background of trees, houses, or herbage. The simplest plan of securing a proper background is to take a disc of ivory or even of white cardboard, and to blacken one side of it. The black paint which is used for this purpose must be without gloss, and have what is called a “dead” surface. Ink answers very well for the purpose, and so does ivory-black; but Indian ink is too glossy to be serviceable.

To procure specimens from the water is a matter of some difficulty if managed badly, but easy enough when the collector knows his business. It is of course needful to attach the collecting vessel to the end of a rod, and to plunge it into the spots which look most favorable. Now even so simple a matter as this requires some little care, if the young microscopist really wishes to obtain the best specimens. A common walking-stick will answer most purposes; but the most efficient rod for the purpose is one of the common walking-stick fishing-rods without the top joint, as it can be carried without attracting attention, and can be lengthened at will by adding the different joints.

Many methods have been proposed by which the vessel is to be attached to the rod; but that which I am about to describe is certainly the simplest and most effective that I have tried. Get a piece of gutta-percha tubing, just large enough to be slipped on the end of the rod or stick; mark off an inch or so, and cut the tube nearly through, then cut it away longitudinally, so that a long tongue of gutta-percha is left, and the instrument is completed.

Its application is as simple as its structure. Bend the tongue over, so as to form a loop, and push the end through the short tube. Slip the neck of the bottle into the loop, and draw the tongue until it is tolerably tight. Push the end of the stick into the tube, taking care to hold the tongue firmly in its place, and the vessel will then be fastened at right angles to the stick.

The method of collecting by means of this instrument is as follows: Immerse the vessel in the water, with the mouth downwards, so that no water may enter. Push it gently towards the spot which is to be investigated, move it about a little, so as to cause a disturbance, and then turn the vessel with its mouth upwards. Water will instantly rush in, carrying with it the objects which are to be examined. The contents of the vessel may then be

transferred to the large bottle, and another dip made. Confervoid growths, especially those which accumulate in a kind of scum on the surface, should be obtained very quietly, without previous disturbance of the water.

After the pond, or stream, or ditch has been well searched, the bottle should be roughly examined, by means of a pocket-lens, and the contents sorted into the smaller tubes, as has already been mentioned. This precaution is especially needful when any of the minute crustacea called Entomostraca are captured, as they are most voracious beings, and will make sad havoc among other specimens, unless they are placed in separate bottles. They are mostly large enough to be detected with the naked eye, and look something like little fleas as they move along.

As the Entomostraca cast their shells repeatedly during their lives, some species performing this operation every two days, a beautiful series of objects can be obtained by gathering the cast shells and preparing them for the microscope, according to the directions that will be found in the following pages. These shells are peculiarly valuable, as they retain the chief external characteristics of the creature to which they belonged, the limbs, plumes, and even the delicate bristles being preserved entire. It is in the power of the microscopist to retard or hasten the change of shell, heat and light aiding development, and cold and darkness retarding it. The remarkable “ephippium,” or saddle, which is found on the backs of the Daphnia, the Moina, and other Entomostraca, and which is used as a receptacle for eggs, should be searched for and preserved.

A very thin and a very flat bottle is a most useful assistance in detecting the character of any unknown object, especially if it be living. Such a bottle may easily be made by heating one of the small test tubes in the spirit lamp until it is of a glowing red heat, and then pressing the sides together. Some little neatness is required in this process, as an unskillful operator is apt to press the sides unequally, and to leave a bulging projection at the end.

THE COMPOUND MICROSCOPE.

We have already described the simpler forms of magnifying instruments, together with the best method of using them. We now purpose to describe the more complicated instrument called the compound microscope, and hints will be given as to the best method of making preparations for it.

The great distinction between the simple and compound microscope is, that whereas the former instrument magnifies the object, the latter magnifies the magnified image of the object. In the least elaborate form of this instrument there are two glasses, one at each end of a tube, the small glass magnifying the object, and being therefore called the “object-glass,” while the other, which magnifies the image of the object, is placed next to the eye, and is therefore termed the “eye-glass.” In practice, however, this arrangement is found to be so extremely defective, that the instrument was quite useless except as an experimental toy; for the two enemies of the optician, chromatic and spherical aberration, prevailed so exceedingly, that every object appeared as if surrounded with prismatic colors, and every line was blurred and indistinct.

In this uncertain state the compound microscope remained for many years, its superb capabilities being scarcely recognized. The chief fault was thought to be in the material of which the object-glass was made, and for a long series of years all experiments were conducted with a view to an improvement in this respect. When, however, the diamond had been employed as an object-glass, and had failed equally with those of less costly material, attention was directed to the right point—namely, the arrangement of the different glasses,—and at length opticians succeeded in obtaining a pitch of excellence which can be almost termed perfection. It would be impossible to describe the method which is employed for this purpose, and it must suffice to say that the principle is that of playing off one defect against another, and so making them mutually correct their errors.

The magnifying powers of the compound microscope can be very great, and it is therefore necessary that extreme care should be taken in its manipulation. It will be possible for a clumsy person to do more damage to a good instrument in three minutes than can be repaired in as many weeks.

Before proceeding to the management of the microscope and the construction of the “slides,” we will briefly describe one or two chief forms of the compound microscope.

The simplest form of the compound microscope, as at present made, consists of a stand and a sliding tube, in which are set the glasses which magnify the object and its image. At the top is the tube, which is capable of being slid up and down in the shoulder of the stand, so as to obtain the

proper focus. Above is seen the eye-glass; and the object-glass is shown at the bottom of the tube. Below the object-glass is the “stage” on which the object to be magnified is laid; and lowest of all is a mirror, which serves to reflect the light upwards through the object, and which can be turned by means of the knobs at the sides. The object-glass is composed of two pieces, which can readily be separated. If both are used, sufficient magnifying power is gained to show the scales on a butterfly’s wing and similar minute objects; while, if one is removed, the object is not magnified to so great an extent, but a larger portion can be seen, and the definition is clearer. The cost of this instrument, together with a few accessories, is about \$2.50.

The proper light is our next point, and upon it rests the chief beauty of the effect. The light which will suit one object will not suit another, and even the same object should be examined under every variety of light. Some objects are best shown when the light is thrown *upon* them from above, and others when it is thrown *through* them from below. Again, the direction of light is of vast importance; for it will easily be seen that an oblique light will exhibit minute projections by throwing a shadow on one side and brilliancy on the other, while a vertical illumination would fail to show them. On the same principle, one object will be shown better with the light in front, and another when it is on one side.

One of the most effective means of attaining this object is by using the “bull’s-eye condenser,” which is sometimes fixed to the stage, but is usually detached. As the upright stem is telescopic, the glass can be raised to a considerable height, while the joint and sliding-rod permit the lens to be applied at any angle which promises the most brilliant light.

As for the kind of light that is employed, there is nothing which equals that of a white cloud; but such clouds are rare, and are at the best extremely transient, and can only be seen by day, various artificial methods of illumination have been invented. Novices generally think that when the sky is bright and blue they will be very successful in their illumination, and feel grievously disappointed at finding that they obtained much more light from the clouds, whose disappearance they had anxiously been watching. Finding that the blue sky gives scarcely any light at all, they rush to the other extreme, turn the mirror towards the sun, and pour such a blaze of light upon the object, that the eye is blinded by the scintillating refulgence, and

the object is often injured because the mirror is capable of reflecting heat as well as light.

In the daytime there is nothing better than the “white-cloud illuminator,” which is made easily enough by means of plaster of Paris. A sheet of thin white paper fastened against a window-pane is also useful; and the simple plan of dabbing the glass with putty will have a beneficial effect in softening the light, when the window has a southern aspect. In default of these conveniences, it will be often sufficient to fix a piece of white letter paper over the mirror, or even to dull its surface with wax. At all events, he who aspires to be a true microscopist must be ready with expedients, and if he finds himself in a difficulty, he must summarily invent a method of obviating it.

At night a lamp is necessary; candles are useless, because they have two faults—they flicker, and they become lower as they burn. The latter defect can be cured by using a candle-lamp, but no arrangement will cure the flame of flickering; it is peculiarly trying to the eyes, and destructive of accurate definition. An ordinary moderator lamp answers pretty well, and a small one is even better for the microscopist than one of large dimensions. The chief drawback to the moderator lamp is that the flame cannot be elevated or lowered, so that the only way to procure a light at a higher elevation is to stand the lamp on a block of wood or a book. Small lamps are, however, made expressly for the microscope, and, if possible, should be procured, and used for no other purpose, and intrusted to no other hands.

If you want a really brilliant, clear, white light, you must trim the lamp yourself. A small piece of pale blue or neutral-tint glass interposed between the lamp and the microscope has a wonderful effect in diminishing the yellow hue which belongs more or less to all artificial lights which are produced by the combustion of oil or fat. We have no doubt but that in a few years we shall be rid of the clumsy and dirty machines that we call lamps, and have substituted for them the pure brilliancy of the electric light.

Whatever lamp you use, a shade is absolutely necessary, in order to defend the eyes. Let me here warn my young readers that they cannot be too careful of their eyes. In the exuberance of youthful strength and health we are too apt to treat our eyes as unceremoniously as our digestion, and in later years we awake to unavailing repentance.

Another point which calls for extreme attention is the perfect cleanliness of the glasses. It is astonishing how a tiny dust-mote, or the least condensation of damp, will diminish the powers of the microscope, and how often the instrument is blamed for indistinctness when the real fault lies in the carelessness of the operator. Even when the greatest care is taken, dust is sure to settle on the glasses, especially on the eye-piece, and before using the microscope the glasses ought to be carefully examined. Never wipe them with an ordinary handkerchief, but get a piece of new wash-leather; beat it well until no dust issues from it, and then put it into a box with a tightly-fitting cover. Use this, and nothing else, for cleaning the glasses, and you will avoid those horrid scratches with which the eye-glass and object-glass of careless operators are always disfigured.

Moisture is very apt to condense on the glasses and to ruin their clearness. If the microscope be brought from a cold into a warm room, the glasses will be instantly covered with moisture, just as the outside of a tumbler of cold water is always covered with fine dew when brought into a warm room. The microscope should therefore be kept at least an hour in the room wherein it is to be used, so that the instrument and the atmosphere may be of the same temperature. You should make the microscope a trifle warmer than the surrounding atmosphere, and so avoid all danger of condensation. When changing the object glass or eye-piece, always keep the hand as far away from the glass as possible, and manipulate with the tip of the forefinger and thumb. The human skin always gives out so much exhalation, that even when the hand is cold the glasses will be dimmed; and it is a peculiarity of such moisture, that it adheres to the glasses with great pertinacity, and does not evaporate like the dew which is condensed from the atmosphere.

In order to insure perfect success in this important particular, the young microscopist will do well to get the optician from whom he purchased his instrument to explain its construction, and to give him a lesson or two in the art of taking it to pieces and putting it together again; for unless each glass can be separately cleaned, no one can be quite sure that the instrument will perform as it ought to do. The best method of ascertaining whether it is quite clean is to throw the light upwards by means of the mirror, and then to turn the eye-piece slowly round. If any dust or moisture has collected either upon the eye-glass or the "field-glass," which forms the second lens of the

eye-piece, it will be immediately detected. Turning the object-glass will in a similar manner detect impurities upon its surface.

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